

The Grand Meson

Shuhei Ihara

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Abstract

This paper states a precise operational thesis: the Grand Meson is the update procedure that tests how far the meson concept can be stretched after the six quark flavors are known, while still recovering the older mesonic face without discontinuity. The starting point is the General Meson: a two-face Yukawa-style mediator read, by convention, through the modern light pair $u \leftrightarrow d$. Given the external six-flavor Standard Model input F_6 , the Grand Meson completes the two-face object to a three-axis, six-face skeleton together with a 36 oriented $q\bar{r}$ -channel index body. The extension is conservative: the original two-face meson is recovered by projection, while the new faces and body channels are resolved as open, inactive, or active instead of being left outside the channel basis. More precisely, the General Meson is the substructure of the Grand Meson in which every non-UD face and every non-UD body channel is buried as a closed inactive residual package. The observed three quark generations are represented as internal Grand Meson directions. Backward compatibility is surjective, but downgrading with unclosed residue is blocked as a Grand Meson contradiction. In this paper, “generalization” means conservative channel-completion with exact recovery, not replacement of QCD; “admissible” means closed under the stated mesonic channels; and “fission” means a contradictory downgrade pattern of residual channels, not nuclear fission. The central claim is therefore strong but bounded: for the present six-flavor quark sector, the Grand Meson is an update manual and a continuity test for mesonic elasticity and robustness.

1 Opening

A meson is usually approached through its physical role: it mediates an interaction. In the present paper the same word is first read through a paired reconstruction. In the present reconstruction, the General Meson is represented by two complementary light-flavor faces. In the Yukawa-facing reconstruction used here, that channel pair is

$$u \leftrightarrow d.$$

This is not a quark–quark bound state. It is the light-flavor channel face from which the QCD readout later forms $q\bar{r}$ channels such as $u\bar{d}$, $d\bar{u}$, $u\bar{u}$, and $d\bar{d}$. Yukawa did not formulate his meson with quark flavors; the notation is the modern light-flavor pair through which this paper rereads the proton–neutron mediation problem.

The former working nickname “Ethon” is retired here. The technical object of this paper is the Grand Meson. The name is structural, not honorific: “grand” means that, after the six-flavor Standard Model input is supplied, a two-face light-pair reconstruction is embedded into the admissible three-axis bookkeeping case.

Convention 1 (Yukawa-facing reconstruction). The implication

$$\text{Yukawa mediator} \rightsquigarrow A_Y = \{u, d\}$$

is not a historical or physical derivation. It is the convention used in this paper: the Yukawa mediator problem is read through the modern light flavor pair u, d . All claims about the General Meson begin only after this reconstruction has been fixed.

2 Operational Thesis

The Grand Meson is not introduced as a new bound state and not as a replacement for the standard quark–antiquark meson. It is introduced as a channel-completion operation on the mesonic flavor sector. Its purpose is to make the following claim explicit:

$$\text{mesonic elasticity} = \text{extension to six flavor faces without loss of recovery.}$$

The corresponding robustness condition is exact projection back to the older face:

$$\begin{aligned} \text{mesonic robustness} &= (p_{\text{ud}}^{\text{raw}}(g) = m) \\ &\quad \wedge \text{ClosedResidue}_d^{\text{face}}(R_{\perp}^{\text{face}}(g)) \\ &\quad \wedge \text{ClosedResidue}_U^{\text{body}}(R_{\perp}^{\text{body}}(G)). \end{aligned}$$

Thus the paper is an update manual for mesonic channel operations and a verification of operational continuity. A Grand Meson statement is well formed only when the older General Meson face is recovered and every residual channel is closed.

Remark 2 (Historical status). The pair $u \leftrightarrow d$ is not claimed to be Yukawa’s original notation. It is a modern light-flavor reconstruction of the mediation problem that Yukawa isolated. The historical input is the mediator problem; the Grand Meson construction is the present six-flavor completion procedure.

3 Standard Mesonic Channel Data

Let the six known quark flavors be

$$F_6 = \{u, d, c, s, t, b\}.$$

Convention 3 (Six-flavor input). The passage from the General Meson to the Grand Meson is not an internal theorem of the two-face object. It is the completion problem obtained after the six known quark flavors F_6 are supplied as external Standard Model input:

$$\text{Grand} = \text{Completion}_{F_6}(\text{Gen}).$$

The symbol Completion_{F_6} is only announced here; it is defined below by the body zero extension z^{body} , the missing-channel completion $\mathcal{C}_{\text{GM}}$, and the skeleton map skel . Thus the additional c, s, t, b directions are not inferred from the Yukawa-facing face alone; they are admitted because the target channel basis is the present six-flavor mesonic sector.

The physical body of a mesonic channel is not a single flavor face but an oriented quark–antiquark channel

$$(q, \bar{r}) \in F_6 \times \overline{F_6}.$$

For a spin-parity sector σ , let Γ_{σ} be the corresponding Dirac structure and define the standard color-singlet interpolating bilinear

$$O_{q\bar{r}}^{\Gamma_{\sigma}}(x) := \sum_{a=1}^3 \bar{r}_a(x) \Gamma_{\sigma} q^a(x).$$

When the spin-parity sector is fixed or irrelevant to the displayed formula, write $O_{q\bar{r}}$ for $O_{q\bar{r}}^{\Gamma_{\sigma}}$. Let Q_q be the electric charge of q , and let $e_q \in \mathbb{Z}^{F_6}$ be the flavor unit at q . For a spin-parity sector σ , define the QCD quantum fingerprint

$$\nu_{\text{QCD}}(q, \bar{r}; \sigma) := (\sigma, Q_q - Q_r, e_q - e_r, B_{\text{ary}} = 0, \mathbf{1}_c).$$

Here B_{ary} denotes baryon number and is not the bottom-flavor label b .

Definition 4 (Mesonic polymer package). For $q, r \in F_6$, a mesonic polymer is the channel package

$$P_{q\bar{r}} := (O_{q\bar{r}}^{\Gamma_\sigma}, J_\sigma^{q\bar{r}}, \nu_{\text{QCD}}(q, \bar{r}; \sigma), C_{q\bar{r}, \bullet}, K_{\text{QCD}}(q\bar{r})).$$

Here $J_\sigma^{q\bar{r}}$ is the source coefficient for the Dirac/spin-parity sector, $C_{q\bar{r}, \bullet}$ denotes the corresponding correlator family, and K_{QCD} denotes the ordinary QCD conservation data. Set

$$\text{PrePol}_{q\bar{r}} := \{ P_{q\bar{r}} \mid P_{q\bar{r}} \text{ is a well-typed package of the displayed form} \}.$$

$$\text{PrePol} := \bigsqcup_{q, r \in F_6} \text{PrePol}_{q\bar{r}}.$$

$$\text{Pol}_{q\bar{r}} := \{ P \in \text{PrePol}_{q\bar{r}} \mid \text{Cl}(P) = \text{Cl}_{\text{GM}} \}.$$

$$\text{Pol} := \bigsqcup_{q, r \in F_6} \text{Pol}_{q\bar{r}}.$$

Thus PrePol is the inspection space and Pol is the closed subspace. The closure predicates below may be evaluated on PrePol ; only packages passing them inhabit Pol .

Definition 5 (Grand Meson closure suite). Let

$$\begin{aligned} \text{Cl}_{\text{QCD}} &:= \{\text{Color}_1, \text{Lorentz}, \text{Charge}, \text{Flavor}\}, \\ \text{Cl}_{\text{weak}} &:= \{\text{CKMCl}, \text{TopPkg}\}, \\ \text{Cl}_{\text{readout}} &:= \{\text{Threshold}, \text{Observable}\}. \end{aligned}$$

The Grand Meson closure suite is the layered union

$$\text{Cl}_{\text{GM}} := \text{Cl}_{\text{QCD}} \cup \text{Cl}_{\text{weak}} \cup \text{Cl}_{\text{readout}}.$$

For a channel package P , $\text{Cl}(P) \subseteq \text{Cl}_{\text{GM}}$ is the set of closure tests satisfied by P . A package is Grand-closed when

$$\text{Cl}(P) = \text{Cl}_{\text{GM}}.$$

The entries of Cl_{GM} are predicates on a package:

$$\begin{aligned} &\text{Color}_1(P), \text{Lorentz}(P), \text{Charge}(P), \text{Flavor}(P), \\ &\text{CKMCl}(P), \text{TopPkg}(P), \text{Threshold}(P), \text{Observable}(P). \end{aligned}$$

For residual closure we use typed predicates rather than one untyped set:

$$\text{RCl}_{\text{face}}(d) \subseteq F_6, \quad \text{RCl}_{\text{body}}(U) \subseteq F_6 \times \overline{F_6}.$$

Let ClosureObjects be the typed disjoint union of closure witnesses used in this paper: QCD closure witnesses, weak-current closure witnesses, top-exit witnesses, threshold witnesses, observable-readout witnesses, and residual-closure records. The channel residual-closure assignment type is

$$\text{ChResType} := \{ \mathcal{R}_{\text{ch}} \mid \mathcal{R}_{\text{ch}}(q\bar{r}) \subseteq \text{ClosureObjects} \text{ for all } q, r \in F_6 \}.$$

Here d is a face downgrade symbol and U is a body transition symbol. Thus $\text{Cl}(P)$ is a test-suite value of a package, while RCl_{face} , RCl_{body} , and ChResType are residual-closure data of different types. The QCD layer contains color singletness, Lorentz covariance, electric-charge conservation, and flavor bookkeeping. The weak layer contains charged-current CKM closure and top-axis weak-virtual closure. The readout layer contains threshold or decay closure and observable readout. The paper never treats CKM closure as a color-QCD closure; it is an

additional weak-current closure condition carried by the Grand Meson package. The weak predicates are conditional predicates:

$$\text{CKMCl}(P) \text{ is vacuous unless } P \text{ uses a weak charged current,}$$

and

$$\text{TopPkg}(P_{q\bar{r}}) \text{ is vacuous unless } q = t \text{ or } r = t.$$

For a top-bearing package, $\text{TopPkg}(P_{q\bar{r}})$ is satisfied only when the weak-virtual top-exit closure defined below is supplied.

Definition 6 (Typed Grand Meson contradiction). A Grand Meson contradiction is not a contradiction of QCD. It is the impossibility of inhabiting the declared Grand Meson readout or transition space. We use the typed family

$$\text{Contr}_{\text{down}}(d), \quad \text{Contr}_{\text{top}}(P_{q\bar{r}}; \mathcal{R}_{\text{ch}}), \quad \text{Contr}_{\bar{F}}(O_{q\bar{r}} \equiv O_{s\bar{t}}), \quad \text{Contr}_{\text{tr}}(U).$$

The shorthand Contr_{GM} refers to this disjoint family only when the input type is clear from context.

Definition 7 (Inactive residual package). Let

$$\text{Prov}_0 := \{\text{selection, kinematic, coupling, sensitivity, integrated, buried, fuse}\}.$$

For each oriented channel $q\bar{r}$, define a closed inactive package

$$\mathbf{0}_{q\bar{r}}[\rho], \quad \rho \in \text{Prov}_0.$$

It is typed by the channel $q\bar{r}$, carries zero source and zero ordinary readout contribution, and satisfies the Grand Meson closure suite as an inactive residual package:

$$\text{Cl}(\mathbf{0}_{q\bar{r}}[\rho]) = \text{Cl}_{\text{GM}}.$$

Set

$$\text{Pol}_{q\bar{r}}^{\text{res}} := \text{Pol}_{q\bar{r}} \cup \{ \mathbf{0}_{q\bar{r}}[\rho] \mid \rho \in \text{Prov}_0 \}, \quad \text{Pol}^{\text{res}} := \bigsqcup_{q,r \in F_6} \text{Pol}_{q\bar{r}}^{\text{res}}.$$

The corresponding pre-closure residual type is

$$\text{PrePol}_{q\bar{r}}^{\text{res}} := \text{PrePol}_{q\bar{r}} \cup \{ \mathbf{0}_{q\bar{r}}[\rho] \mid \rho \in \text{Prov}_0 \}.$$

Definition 8 (Face package). A flavor face is not itself an oriented mesonic channel. It is the full family of oriented channels incident to that face relative to a chosen flavor domain $S \subseteq F_6$:

$$\text{Face}_{q|S}^{\text{res}} := \prod_{r \in S} \text{PrePol}_{q\bar{r}}^{\text{res}} \times \prod_{s \in S} \text{PrePol}_{s\bar{q}}^{\text{res}}.$$

Write $\text{Face}_q^{\text{res}} := \text{Face}_{q|F_6}^{\text{res}}$ for the full Grand Meson face. The closed inactive face package is therefore

$$\mathbf{0}_{q|S}[\rho] := ((\mathbf{0}_{q\bar{r}}[\rho])_{r \in S}, (\mathbf{0}_{s\bar{q}}[\rho])_{s \in S}) \in \text{Face}_{q|S}^{\text{res}}.$$

Write $\mathbf{0}_q[\rho] := \mathbf{0}_{q|F_6}[\rho]$. Set

$$\text{Face}^{\text{res}} := \prod_{q \in F_6} \text{Face}_q^{\text{res}}.$$

4 The General Meson

Let the Yukawa-facing readable pair be

$$A_Y = \{u, d\}.$$

The smallest face-level implementation may be generated from

$$\mathbb{P}_3 = \{-\mathbf{1}, \mathbf{0}, +\mathbf{1}\},$$

but no physical charge, amplitude, or particle identity is assigned to these pseudo-values before a $q\bar{r}$ -channel package is supplied.

Remark 9 (Role of Pol). Pol denotes the active closed mesonic package space above, and Pol^{res} adds closed inactive residual packages. Containers such as faces and bodies use $\text{PrePol}^{\text{res}}$ so that closure tests can still reject an open active package at readout time. The six-face notation is therefore a skeleton notation built from $\text{Face}_q^{\text{res}}$. The physical mesonic body lives over $F_6 \times \overline{F_6}$.

Definition 10 (General Meson). A raw General Meson is a two-face mesonic skeleton

$$m \in \text{Gen}^{\text{raw}} := \prod_{q \in A_Y} \text{Face}_{q|A_Y}^{\text{res}}.$$

Its inspectable oriented UD body is

$$\text{PreBody}_Y := \prod_{q, r \in A_Y} \text{PrePol}_{q\bar{r}}^{\text{res}}.$$

Its closed oriented UD body is

$$\text{Body}_Y^{\text{cl}} := \{ M \in \text{PreBody}_Y \mid \text{Inact}(M(q, \bar{r})) \text{ or } \text{Cl}(M(q, \bar{r})) = \text{Cl}_{\text{GM}} \text{ for all } q, r \in A_Y \}.$$

Define the UD skeleton map

$$\text{skel}_Y : \text{PreBody}_Y \longrightarrow \text{Gen}^{\text{raw}}$$

by

$$\text{skel}_Y(M)(q) := ((M(q, \bar{r}))_{r \in A_Y}, (M(s, \bar{q}))_{s \in A_Y}).$$

The glued General Meson is

$$\text{Gen}_{\text{comp}} := \text{Im}(\text{skel}_Y : \text{Body}_Y^{\text{cl}} \rightarrow \text{Gen}^{\text{raw}}) \subseteq \text{Gen}^{\text{raw}}.$$

The map skel_Y is injective, since each $M(q, \bar{r})$ occurs in the outgoing family of the q -face. Hence a glued General Meson has a unique body, denoted $\text{body}_Y(m)$. From this point onward, unqualified Gen means Gen_{comp} . The faces u and d are the two light-flavor faces of the Yukawa-style mesonic channel. They are face packages, not two constituents of a physical meson. The physical body supplied by those faces is the four oriented UD channels

$$u\bar{u}, \quad u\bar{d}, \quad d\bar{u}, \quad d\bar{d}.$$

This definition deliberately keeps the Yukawa-style mediator idea narrow. It does not say that a meson is universal. It says only that the basic mesonic act is readable mediation across the u, d light pair. In that restricted sense, the General Meson is a deliberately narrow readout for the nuclear-force problem: the proton–neutron transition axis is specified by the reconstruction convention. The light-pair configuration is not by itself a derivation of the Yukawa potential. That recovery belongs to the low-energy pion channel after the QCD readout has supplied $q\bar{r}$ operators, quantum numbers, and observables.

5 The Hidden Cost of Missing Channels

The decisive distinction is between an inactive channel and a missing channel. An inactive channel belongs to the channel basis and carries no active contribution. A missing channel has not entered the channel basis.

$\mathbf{0}$ = resolved inactive channel, $\perp_{q\bar{r}}$ = missing oriented channel, $\perp_q$ = missing face.

Definition 11 (Inactive-channel provenance). The symbol $\mathbf{0}$ is a formal null, not a universal physical zero. When physical input is available, an inactive channel may be refined by a provenance tag

$$\mathbf{0}[\rho], \quad \rho \in \{\text{selection, kinematic, coupling, sensitivity, integrated}\}.$$

Thus a selection-blocked channel, an energetically unreachable channel, a zero coupling, an experimentally unresolved signal, and an integrated-out contribution are not collapsed into the same physical claim.

Definition 12 (Missing-channel lift). The missing-channel lift of a General Meson $m \in \text{Gen}$ is

$$L_\perp(m) \in \prod_{q \in F_6} \text{Face}_{q|F_6}^\perp,$$

where

$$\text{Face}_{q|S}^\perp := \prod_{r \in S} (\text{PrePol}_{q\bar{r}}^{\text{res}} \cup \{\perp_{q\bar{r}}\}) \times \prod_{s \in S} (\text{PrePol}_{s\bar{q}}^{\text{res}} \cup \{\perp_{s\bar{q}}\}).$$

For $q \in A_Y$, $L_\perp(m)(q)$ agrees with $m(q)$ on the incident UD slots and assigns the appropriate missing symbols to every incident slot whose other endpoint lies outside A_Y . For $q \notin A_Y$, every incident slot of $L_\perp(m)(q)$ is missing; this face is written $\perp_{q|F_6}$.

Definition 13 (Componentwise missing-channel completion). Missing channels do not become inactive by inference. They become inactive only by the Grand completion operator

$$\mathcal{C}_{\text{GM}}.$$

On an already supplied oriented package, it is the identity:

$$\mathcal{C}_{\text{GM}}(P) = P \quad (P \in \text{PrePol}_{q\bar{r}}^{\text{res}}).$$

On an oriented missing channel it is defined by

$$\mathcal{C}_{\text{GM}}(\perp_{q\bar{r}}) := \mathbf{0}_{q\bar{r}}[\text{buried}],$$

and on a missing face carrying its relative domain $S \subseteq F_6$, by

$$\mathcal{C}_{\text{GM}}(\perp_{q|S}) := \mathbf{0}_{q|S}[\text{buried}].$$

The operator extends componentwise to partial face skeletons and partial body skeletons: it fixes every supplied package and sends every missing entry to the corresponding buried inactive package. Therefore $\mathcal{C}_{\text{GM}}(L_\perp(m))$ means componentwise completion of all missing entries of $L_\perp(m)$ while leaving the already supplied UD data unchanged. Thus

$$\perp_{q\bar{r}} \not\equiv \mathbf{0}_{q\bar{r}}[\text{buried}], \quad \mathcal{C}_{\text{GM}}(\perp_{q\bar{r}}) = \mathbf{0}_{q\bar{r}}[\text{buried}].$$

The first formula preserves the distinction between absent and inactive; the second formula names the completion rule that upgrades an absent channel into a present, closed, inactive residual package.

Thus the remaining axes are not false, not zero, and not disproved. They are absent from the original face basis. The two-face theory remains locally correct while it cannot close propagation that travels through the missing directions.

Remark 14 (Physical reading). The missing faces are not a defect in the original General Meson. They are the price of a successful restriction. A narrow mediator may solve its local problem precisely because it does not carry the full burden of every other axis.

6 The Grand Meson Face Skeleton

Given the external six-flavor Standard Model input F_6 , the first layer of the Grand Meson is obtained by completing missing flavor faces into resolved flavor faces. This is the raw face skeleton. It is not yet the full mesonic operator body; physical mesons enter through the oriented $q\bar{r}$ -channel completion defined later.

Definition 15 (Grand Meson face skeleton). A Grand Meson first appears as a raw six-face mesonic skeleton

$$g \in \mathbf{Grand}^{\text{raw}} := \prod_{q \in F_6} \mathbf{Face}_q^{\text{res}}.$$

This raw product is larger than the body-compatible Grand Meson defined later: it has not yet imposed that the same $q\bar{r}$ package agrees when read from the q -face and from the r -face. The six faces are grouped into three readable axes:

$$u \leftrightarrow d, \quad c \leftrightarrow s, \quad t \leftrightarrow b.$$

These axes are internal flavor-generation axes. They are not spatial directions, and they are not quark–quark bound states. Physical mesons enter through oriented $q\bar{r}$ channels in the QCD readout.

In this notation the Grand Meson is the admissible $n = 3$ object:

$$\mathbf{Grand}^{\text{raw}} = \prod_{q \in F_6} \mathbf{Face}_q^{\text{res}} \quad (n = 3).$$

The adjective “admissible” means that all three axes belong to the channel basis after the external six-flavor input has been supplied. It does not mean that every transition is allowed, nor that every physical theory has been solved.

Definition 16 (Channel support). For an oriented package, write

$$\text{Inact}(P_{q\bar{r}}) \iff P_{q\bar{r}} = \mathbf{0}_{q\bar{r}}[\rho] \text{ for some } \rho \in \text{Prov}_0.$$

Also write

$$\text{Buried}(P_{q\bar{r}}) \iff P_{q\bar{r}} = \mathbf{0}_{q\bar{r}}[\text{buried}].$$

For a face package $f_q \in \mathbf{Face}_q^{\text{res}}$, write $\text{Inact}(f_q)$ when every incident oriented component of f_q is inactive, allowing different provenance tags on different incident channels. The predicate is overloaded on face packages by

$$\text{Buried}(f_q) \iff \text{every incident component of } f_q \text{ is } \mathbf{0}[\text{buried}] \text{ in its own channel type.}$$

For $g \in \mathbf{Grand}^{\text{raw}}$, define the face support by

$$\text{supp}_{\text{face}}(g) = \{q \in F_6 \mid \neg \text{Inact}(g(q))\}.$$

A Grand Meson may have inactive channels, but inactivity is resolved by channelwise residual packages, not by $\perp$.

7 Conservative Recovery

The Grand Meson must not destroy the General Meson. It must contain it as the two-channel face that was already working.

Definition 17 (Raw projection to the General Meson). Define the raw u, d-projection

$$p_{\text{ud}}^{\text{raw}} : \text{Grand}^{\text{raw}} \rightarrow \text{Gen}^{\text{raw}}$$

by

$$p_{\text{ud}}^{\text{raw}}(g)(q) = g(q)|_{A_Y} \quad (q \in A_Y),$$

where $g(q)|_{A_Y}$ forgets every incident channel outside $A_Y \times \overline{A_Y}$.

Proposition 18 (Conservative recovery). *Every raw General Meson is recovered from any raw Grand Meson whose restricted u, d-faces agree with it. In particular, if $g(q)|_{A_Y} = m(q)$ for $q \in A_Y$, then*

$$p_{\text{ud}}^{\text{raw}}(g) = m.$$

Proof. The restriction $p_{\text{ud}}^{\text{raw}}$ reads exactly the two light-face packages. Therefore agreement on those two faces is sufficient and necessary for the stated projection recovery. $\square$

This is the main discipline of the construction. The Grand Meson does not rewrite the older meson. It keeps the working $u \leftrightarrow d$ face and opens two additional quark-pair faces under explicit channel closure.

8 Surjective Backward Compatibility

Backward compatibility must be surjective, not merely suggestive. Every General Meson must be obtainable as the visible face of at least one Grand Meson.

Definition 19 (Zero extension). For $m \in \text{Gen}^{\text{raw}}$, define the raw zero extension

$$z^{\text{raw}}(m) \in \text{Grand}^{\text{raw}}$$

as follows. For $q \in A_Y$, let $\widehat{m}_q \in \text{Face}_q^{\text{res}}$ be the full-face extension that agrees with $m(q)$ on the incident UD channels and assigns $\mathbf{0}_{ab}[\text{buried}]$ to every incident channel outside $A_Y \times \overline{A_Y}$. Then

$$z^{\text{raw}}(m)(u) = \widehat{m}_u, \quad z^{\text{raw}}(m)(d) = \widehat{m}_d,$$

and

$$z^{\text{raw}}(m)(q) = \mathbf{0}_q[\text{buried}] \quad (q \in F_6 \setminus A_Y).$$

Proposition 20 (Surjective backward map). *The projection $p_{\text{ud}}^{\text{raw}} : \text{Grand}^{\text{raw}} \rightarrow \text{Gen}^{\text{raw}}$ satisfies surjective:*

$$p_{\text{ud}}^{\text{raw}} \circ z^{\text{raw}} = \text{id}_{\text{Gen}^{\text{raw}}}.$$

Proof. The zero extension preserves the u, d-face of m and resolves all other channels as inactive. Projecting back to the u, d-face therefore returns exactly m . $\square$

Definition 21 (UD burial normalization). Define the UD-burial subobject

$$\text{Grand}_{\text{ud}}^0 := z^{\text{raw}}(\text{Gen}^{\text{raw}}) = \{ z^{\text{raw}}(m) \mid m \in \text{Gen}^{\text{raw}} \}.$$

The burial normalization

$$\beta_{\text{ud}} : \text{Grand}^{\text{raw}} \longrightarrow \text{Grand}_{\text{ud}}^0$$

is defined by

$$\beta_{\text{ud}}(g) := z^{\text{raw}}(p_{\text{ud}}^{\text{raw}}(g)).$$

Here $\mathbf{0}_q[\text{buried}]$ is a closed inactive residual package, not a deletion and not an absent channel. Other provenance tags describe other inactive mechanisms, but the canonical recovery of $\mathbf{Gen}$ uses the buried tag.

Proposition 22 (General Meson as buried Grand Meson). *Restriction to the UD face gives an isomorphism*

$$\mathbf{Grand}_{\text{ud}}^0 \cong \mathbf{Gen}^{\text{raw}}.$$

More explicitly,

$$p_{\text{ud}}^{\text{raw}}|_{\mathbf{Grand}_{\text{ud}}^0} : \mathbf{Grand}_{\text{ud}}^0 \rightarrow \mathbf{Gen}^{\text{raw}}$$

has inverse $z^{\text{raw}} : \mathbf{Gen}^{\text{raw}} \rightarrow \mathbf{Grand}_{\text{ud}}^0$.

Proof. By definition, every $h \in \mathbf{Grand}_{\text{ud}}^0$ has the form $h = z^{\text{raw}}(m)$ for a unique $m \in \mathbf{Gen}^{\text{raw}}$. Conversely, $z^{\text{raw}}(m)$ preserves the UD-restricted face data of m and fills every incident non-UD channel with a closed inactive package. Hence

$$p_{\text{ud}}^{\text{raw}} \circ z^{\text{raw}} = \text{id}_{\mathbf{Gen}^{\text{raw}}}, \quad z^{\text{raw}} \circ p_{\text{ud}}^{\text{raw}} = \text{id}_{\mathbf{Grand}_{\text{ud}}^0}.$$

□

Surjectivity gives power: a Grand Meson can always be normalized to an ordinary General Meson by burying the non-UD faces as closed inactive packages. The same operation also carries responsibility, because active non-UD structure must be closed as residue rather than ignored.

Axiom 23 (Grand Meson downgrade admissibility). Recoverability is not the same as downgrade admissibility:

$$p_{\text{ud}}^{\text{raw}}(g) = m \not\Rightarrow d \in \text{Dow}_{\text{GM}}(g, m).$$

A proposed downgrade $d : g \Downarrow m$ is admissible only when it both recovers the General Meson face and supplies closed residual witnesses for the non-UD data:

$$d \in \text{Dow}_{\text{GM}}(g, m) \iff p_{\text{ud}}^{\text{raw}}(g) = m \text{ and } \text{ClosedResidue}_d^{\text{face}}(R_{\perp}^{\text{face}}(g)).$$

Thus residue responsibility is an explicit Grand Meson admissibility rule, not a theorem following from $p_{\text{ud}}^{\text{raw}} \circ z^{\text{raw}} = \text{id}$.

Definition 24 (Downgrade residue and closure). For $g \in \mathbf{Grand}^{\text{raw}}$, define the downgrade residue

$$R_{\perp}^{\text{face}}(g) := g|_{F_6 \setminus A_Y}$$

and its non-buried residue support

$$S_{\perp}^{\text{face}}(g) := \{ q \in F_6 \setminus A_Y \mid \neg \text{Buried}(g(q)) \}.$$

For a proposed downgrade symbol $d : g \Downarrow m$, the predicate

$$\text{ClosedResidue}_d^{\text{face}}(R_{\perp}^{\text{face}}(g))$$

means that d carries a residual witness

$$\chi_d : S_{\perp}^{\text{face}}(g) \longrightarrow \text{RCl}_{\text{face}}(d)$$

and that every face in $S_{\perp}^{\text{face}}(g)$ is sent by the normalization to the closed inactive family $\mathbf{0}_q[\text{buried}]$. Projection alone is therefore not a downgrade; projection plus residual witness is. For a proposed downgrade symbol $d : g \Downarrow m$, define the Grand Meson downgrade class

$$\text{Dow}_{\text{GM}}(g, m) := \{ d : g \Downarrow m \mid p_{\text{ud}}^{\text{raw}}(g) = m \text{ and } \text{ClosedResidue}_d^{\text{face}}(R_{\perp}^{\text{face}}(g)) \}.$$

The contradiction predicate is

$$\text{Contr}_{\text{dow}}(d) := (p_{\text{ud}}^{\text{raw}}(s(d)) = t(d)) \wedge (S_{\perp}^{\text{face}}(s(d)) \neq \emptyset) \wedge \neg \text{ClosedResidue}_d^{\text{face}}(R_{\perp}^{\text{face}}(s(d))).$$

Here $s(d) = g$ and $t(d) = m$ are the source and target carried by the typed downgrade symbol $d : g \Downarrow m$.

Proposition 25 (Robust downgrade block). *If $\text{Contr}_{\text{dow}}(d)$ holds for a proposed downgrade symbol $d : g \Downarrow m$, then*

$$d \notin \text{Dow}_{\text{GM}}(g, m).$$

If every proposed downgrade symbol from g to m is contradictory, then

$$\text{Dow}_{\text{GM}}(g, m) = \emptyset.$$

Proof. Membership in $\text{Dow}_{\text{GM}}(g, m)$ requires the residue predicate

$$\text{ClosedResidue}_d^{\text{face}}(R_{\perp}^{\text{face}}(g)).$$

The predicate $\text{Contr}_{\text{dow}}(d)$ contains its negation. The two conditions cannot hold simultaneously, hence d is not a Grand Meson downgrade. If no proposed symbol satisfies the membership condition, the class is empty. $\square$

Remark 26 (Fusion and fission). Fusion is the closed downgrade: the active extra axes are collapsed into the General Meson face only with an explicit residual channel. Fission is not a second operation. It is the name of the contradictory downgrade pattern in which the same projection is written while the extra axes are discarded as if they had never contributed. In the Grand Meson channel discipline, this pattern is blocked by emptiness of $\text{Dow}_{\text{GM}}(g, m)$. The word does not denote nuclear fission in this paper.

9 QCD Channel-Index Lift

The Grand Meson does not complete QCD dynamics. It performs a conservative channel-index lift of the six-flavor mesonic operator basis: it adds a recoverable generation-axis indexing layer to the standard color-singlet quark–antiquark bilinear. It does not alter QCD color dynamics, does not replace confinement, and does not claim a new bound state by notation alone. Its channel-index lift is

$$O_{q\bar{r}} \mapsto (i, j, O_{q\bar{r}}), \quad q \in A_i, \ r \in A_j,$$

together with the recovery map that forgets (i, j) . Let the three flavor axes be

$$A_1 = \{u, d\}, \quad A_2 = \{c, s\}, \quad A_3 = \{t, b\}.$$

Definition 27 (Grand Meson channel completion). The typed channel-label layer is

$$\text{GM}_{36}^{\text{ch}} := \{ (i, j, P_{q\bar{r}}) \mid q \in A_i, \ r \in A_j, \ P_{q\bar{r}} \in \text{PrePol}_{q\bar{r}}^{\text{res}} \}.$$

The inspectable 36-channel body is the dependent product

$$\text{PreBody}_{36} := \prod_{q, r \in F_6} \text{PrePol}_{q\bar{r}}^{\text{res}}.$$

An element $G \in \text{PreBody}_{36}$ has a value

$$G(q, \bar{r}) \in \text{PrePol}_{q\bar{r}}^{\text{res}}$$

in the correct oriented channel type. The closed 36-channel body is

$$\text{Body}_{36}^{\text{cl}} := \{ G \in \text{PreBody}_{36} \mid \text{Inact}(G(q, \bar{r})) \text{ or } \text{Cl}(G(q, \bar{r})) = \text{Cl}_{\text{GM}} \text{ for all } q, r \in F_6 \}.$$

When the closed body is intended, write

$$\text{Body}_{36} := \text{Body}_{36}^{\text{cl}}, \quad \text{GM}_{36} := \text{Body}_{36}^{\text{cl}}.$$

The six-face skeleton records incident channel families; the $F_6 \times \overline{F_6}$ dependent product is the mesonic channel body. Its channelwise active support is

$$\text{supp}_{\text{ch}}(G) := \{ (q, \bar{r}) \mid \neg \text{Inact}(G(q, \bar{r})) \},$$

and its body-derived face support is the projection

$$\text{supp}_{\text{face}}(G) := \{ q \in F_6 \mid (q, \bar{r}) \in \text{supp}_{\text{ch}}(G) \text{ or } (s, \bar{q}) \in \text{supp}_{\text{ch}}(G) \text{ for some } r, s \in F_6 \}.$$

Definition 28 (Skeleton of a body). Define the skeleton map

$$\text{skel} : \text{PreBody}_{36} \longrightarrow \text{Grand}^{\text{raw}}$$

by

$$\text{skel}(G)(q) := ((G(q, \bar{r}))_{r \in F_6}, (G(s, \bar{q}))_{s \in F_6}) \in \text{Face}_q^{\text{res}}.$$

A pair (g, G) is skeleton–body compatible when

$$g = \text{skel}(G).$$

The map skel is injective, since each body value $G(q, \bar{r})$ appears in the outgoing family of the q -face. Thus a face is never identified with a single $q\bar{r}$ package; it is the incident family extracted from the body.

Definition 29 (Glued Grand Meson skeleton). The inspectable body-compatible skeleton is

$$\text{PreGrand}_{\text{comp}} := \text{Im}(\text{skel} : \text{PreBody}_{36} \rightarrow \text{Grand}^{\text{raw}}).$$

The closed body-compatible Grand Meson skeleton is

$$\text{Grand}_{\text{comp}} := \text{Im}(\text{skel} : \text{Body}_{36}^{\text{cl}} \rightarrow \text{Grand}^{\text{raw}}) \subseteq \text{Grand}^{\text{raw}}.$$

An element $g \in \text{Grand}^{\text{raw}}$ is glued when there exists some $G \in \text{PreBody}_{36}$ such that $g = \text{skel}(G)$. It is a closed Grand Meson skeleton when such a witness may be chosen in $\text{Body}_{36}^{\text{cl}}$. Equivalently, every incident appearance of the same oriented package $P_{q\bar{r}}$ agrees across the two faces that mention it, and closed Grand Mesons additionally have no unclosed active component. From this point onward, unqualified Grand means the closed glued object $\text{Grand}_{\text{comp}}$.

Definition 30 (UD burial on the channel body). Let

$$\text{PreGM}_{36, \text{ud}}^{0, \text{body}} := \{ G \in \text{PreBody}_{36} \mid G(q, \bar{r}) = \mathbf{0}_{q\bar{r}}[\text{buried}] \text{ for every } (q, \bar{r}) \notin A_Y \times \overline{A_Y} \}.$$

Set

$$\text{GM}_{36, \text{ud}}^{0, \text{body}} := \text{PreGM}_{36, \text{ud}}^{0, \text{body}} \cap \text{Body}_{36}^{\text{cl}}.$$

Define the channel-body burial normalization

$$\mathcal{B}_{\text{ud}} : \text{PreBody}_{36} \rightarrow \text{PreGM}_{36,\text{ud}}^{0,\text{body}}$$

by

$$\mathcal{B}_{\text{ud}}(G)(q, \bar{r}) = \begin{cases} G(q, \bar{r}), & (q, \bar{r}) \in A_Y \times \overline{A_Y}, \\ \mathbf{0}_{q\bar{r}}[\text{buried}], & (q, \bar{r}) \notin A_Y \times \overline{A_Y}. \end{cases}$$

Thus every cross-generation or non-UD body channel is retained as a closed inactive residual package, not deleted and not forgotten. The body projection is

$$p_{\text{ud}}^{\text{body}}(G) := G|_{A_Y \times \overline{A_Y}} \in \text{PreBody}_Y.$$

Definition 31 (Body zero extension and F_6 -completion). For $M \in \text{Body}_Y^{\text{cl}}$, define the body zero extension

$$z^{\text{body}}(M) \in \text{Body}_{36}^{\text{cl}}$$

by

$$z^{\text{body}}(M)(q, \bar{r}) = \begin{cases} M(q, \bar{r}), & (q, \bar{r}) \in A_Y \times \overline{A_Y}, \\ \mathcal{C}_{\text{GM}}(\perp_{q\bar{r}}), & (q, \bar{r}) \notin A_Y \times \overline{A_Y}. \end{cases}$$

For $m \in \text{Gen}$, define the announced six-flavor completion by

$$z^{\text{comp}}(m) := \text{Completion}_{F_6}(m) := \text{skel}\left(z^{\text{body}}(\text{body}_Y(m))\right) \in \text{Grand}_{\text{comp}}.$$

Equivalently,

$$z^{\text{comp}}(m) = \text{Completion}_{F_6}(m) = \mathcal{C}_{\text{GM}}(L_{\perp}(m)),$$

where the last expression means: preserve the UD face data and apply $\mathcal{C}_{\text{GM}}$ to every missing non-UD incident channel.

Proposition 32 (Completion identities). *The local completion symbols introduced above are connected by the body witness*

$$z^{\text{comp}}(m) = \text{skel}\left(z^{\text{body}}(\text{body}_Y(m))\right) \in \text{Grand}_{\text{comp}}$$

and the two identities

$$z^{\text{comp}}(m) = \mathcal{C}_{\text{GM}}(L_{\perp}(m))$$

for every glued General Meson $m \in \text{Gen} = \text{Gen}_{\text{comp}}$, and

$$\text{Grand}_{\text{comp}} = \text{Im}(\text{skel} : \text{Body}_{36}^{\text{cl}} \rightarrow \text{Grand}^{\text{raw}}).$$

The first identity says that zero extension is the missing-channel lift followed by buried completion. The second says that a Grand Meson skeleton is closed and glued exactly when it comes from a closed 36-channel body.

Proof. Since m is glued, $M := \text{body}_Y(m)$ exists and satisfies $m = \text{skel}_Y(M)$. The body zero extension $z^{\text{body}}(M)$ preserves the UD body M and sends every non-UD body channel to its buried inactive package. Applying skel gives exactly the face-level zero completion $z^{\text{comp}}(m)$. Hence $z^{\text{comp}}(m)$ has the body witness $z^{\text{body}}(\text{body}_Y(m))$ and lies in $\text{Im}(\text{skel}|_{\text{Body}_{36}^{\text{cl}}}) = \text{Grand}_{\text{comp}}$. The componentwise definition of $\mathcal{C}_{\text{GM}}$ sends each missing non-UD channel of $L_{\perp}(m)$ to the same buried inactive package and leaves the UD data represented by m unchanged; this is the same $z^{\text{comp}}(m)$. The final identity is the definition of $\text{Grand}_{\text{comp}}$. $\square$

Definition 33 (Glued UD projection). Define the glued projection

$$p_{\text{ud}} : \text{Grand}_{\text{comp}} \rightarrow \text{Gen}$$

by

$$p_{\text{ud}}(\text{skel}(G)) := \text{skel}_Y(p_{\text{ud}}^{\text{body}}(G)).$$

This is the restriction of the raw projection to glued skeletons, with the codomain restricted to glued General Mesons.

Definition 34 (Closed UD burial subobject). The closed UD-burial subobject is

$$\text{Grand}_{\text{ud}}^{0,\text{comp}} := z^{\text{comp}}(\text{Gen}) \subseteq \text{Grand}_{\text{comp}}.$$

Proposition 35 (Closed General Meson recovery). *Restriction to the UD face gives an isomorphism*

$$\text{Grand}_{\text{ud}}^{0,\text{comp}} \cong \text{Gen}.$$

More explicitly,

$$p_{\text{ud}}|_{\text{Grand}_{\text{ud}}^{0,\text{comp}}} : \text{Grand}_{\text{ud}}^{0,\text{comp}} \rightarrow \text{Gen}$$

has inverse $z^{\text{comp}} : \text{Gen} \rightarrow \text{Grand}_{\text{ud}}^{0,\text{comp}}$.

Proof. By definition, every element of $\text{Grand}_{\text{ud}}^{0,\text{comp}}$ has the form $z^{\text{comp}}(m)$ for a unique $m \in \text{Gen}$. Since m has a closed UD body witness, $z^{\text{comp}}(m) = \text{skel}(z^{\text{body}}(\text{body}_Y(m)))$ has a closed 36-channel body witness and therefore belongs to $\text{Grand}_{\text{comp}}$. The glued projection returns the UD body witness, hence $p_{\text{ud}}(z^{\text{comp}}(m)) = m$. $\square$

Proposition 36 (Glued backward recovery). *For every $m \in \text{Gen}$,*

$$p_{\text{ud}}(\text{Completion}_{F_6}(m)) = m.$$

Equivalently,

$$p_{\text{ud}} \circ \text{Completion}_{F_6} = \text{id}_{\text{Gen}}.$$

Proof. Let $M = \text{body}_Y(m)$. By definition,

$$\text{Completion}_{F_6}(m) = \text{skel}(z^{\text{body}}(M)).$$

Applying p_{ud} restricts the completed body back to $A_Y \times \overline{A_Y}$, which is exactly M . Applying skel_Y then returns m . $\square$

Definition 37 (Body downgrade residue and closure). For $G \in \text{PreBody}_{36}$, define

$$R_{\perp}^{\text{body}}(G) := G|_{(F_6 \times \overline{F_6}) \setminus (A_Y \times \overline{A_Y})}$$

and

$$S_{\perp}^{\text{body}}(G) := \{ (q, \bar{r}) \notin A_Y \times \overline{A_Y} \mid G(q, \bar{r}) \neq \mathbf{0}_{q\bar{r}}[\text{buried}] \}.$$

For a body downgrade or transition witness U , the predicate

$$\text{ClosedResidue}_U^{\text{body}}(R_{\perp}^{\text{body}}(G))$$

means that U carries a witness

$$\chi_U : S_{\perp}^{\text{body}}(G) \longrightarrow \text{RCl}_{\text{body}}(U)$$

and that $\mathcal{B}_{\text{ud}}(G)$ sends every such channel to $\mathbf{0}_{q\bar{r}}[\text{buried}]$.

Proposition 38 (Body-level General Meson recovery). *Restriction to $A_Y \times \overline{A_Y}$ gives*

$$\mathbf{GM}_{36,\text{ud}}^{0,\text{body}} \cong \mathbf{Body}_Y^{\text{cl}}.$$

Proof. Every channel outside $A_Y \times \overline{A_Y}$ is fixed to the closed inactive package $\mathbf{0}_{q\bar{r}}[\text{buried}]$, so an element of $\mathbf{GM}_{36,\text{ud}}^{0,\text{body}}$ is determined exactly by its four oriented UD channels:

$$u\bar{u}, \quad u\bar{d}, \quad d\bar{u}, \quad d\bar{d}.$$

Conversely, any package assignment on these four channels extends uniquely by buried inactive packages on the remaining 32 channels. $\square$

Proposition 39 (Skeleton–body burial compatibility). *For every $G \in \text{PreBody}_{36}$,*

$$\text{skel}(\mathcal{B}_{\text{ud}}(G)) = \beta_{\text{ud}}(\text{skel}(G)).$$

Thus face-level burial and body-level burial are the same normalization seen from two typed layers.

Proof. For $q \in A_Y$, both sides keep the incident UD family. For $q \notin A_Y$, every incident body channel of $\mathcal{B}_{\text{ud}}(G)$ is a buried inactive package, hence the skeleton is $\mathbf{0}_q[\text{buried}]$. This is exactly the definition of β_{ud} applied to $\text{skel}(G)$. $\square$

Definition 40 (Generation as internal direction). Let

$$G_3 = \{1, 2, 3\}, \quad H_2 = \{+, -\}.$$

Define

$$\Phi : G_3 \times H_2 \longrightarrow F_6$$

by

$$(1, +) \mapsto u, \quad (1, -) \mapsto d, \quad (2, +) \mapsto c, \quad (2, -) \mapsto s, \quad (3, +) \mapsto t, \quad (3, -) \mapsto b.$$

Then

$$A_i = \Phi(\{i\} \times H_2).$$

Proposition 41 (Generation-index decomposition of the Grand Meson skeleton). *The raw six-flavor face layer decomposes as*

$$\mathbf{Grand}^{\text{raw}} = \prod_{q \in F_6} \mathbf{Face}_q^{\text{res}} \cong \prod_{(i,\epsilon) \in G_3 \times H_2} \mathbf{Face}_{\Phi(i,\epsilon)}^{\text{res}} \cong \prod_{i \in G_3} \prod_{\epsilon \in H_2} \mathbf{Face}_{\Phi(i,\epsilon)}^{\text{res}}.$$

The glued Grand Meson $\mathbf{Grand}_{\text{comp}}$ inherits this three-axis indexing through its inclusion into $\mathbf{Grand}^{\text{raw}}$. Thus the three observed quark generations are not derived from the General Meson alone; they enter as the fixed six-flavor input and become internal directions of the completed skeleton.

Definition 42 (Top-axis closure). The third direction

$$A_3 = \{t, b\}$$

is not an ordinary top-meson axis. It is a weak-virtual channel axis. For a channel $q\bar{r}$, define

$$\text{TopCh}(q\bar{r}) \iff (q = t) \text{ or } (r = t).$$

An ordinary mesonic readout is defined only when $\text{TopCh}(q\bar{r})$ is false. If $\text{TopCh}(q\bar{r})$ is true, the channel closes only through a weak-virtual channel closure relative to a channel residual assignment $\mathcal{R}_{\text{ch}} \in \text{ChResType}$:

$$\text{WCl}(q\bar{r}) \subseteq \mathcal{R}_{\text{ch}}(q\bar{r}).$$

The closure object is the weak charged-current exit available before an ordinary top-bearing mesonic readout can be formed. Let

$$d_1 = d, \quad d_2 = s, \quad d_3 = b.$$

$$\text{WCl}_L(t\bar{r}) := \{ t \rightarrow d_j W^+ \text{ with } V_{3j} \mid j = 1, 2, 3 \},$$

and, for an anti-top-bearing channel,

$$\text{WCl}_R(q\bar{t}) := \{ \bar{t} \rightarrow \bar{d}_j W^- \text{ with } \overline{V_{3j}} \mid j = 1, 2, 3 \}.$$

For a general channel, $\text{WCl}(q\bar{r})$ is the union of all available left and right top exits; for $t\bar{t}$, both sides are required. If neither side contains a top label, $\text{WCl}(q\bar{r})$ is empty. Thus the top axis is present in the six-flavor completion, but ordinary mesonic readability is blocked unless the weak-virtual closure is supplied.

Proposition 43 (Unclosed top readout is GM-contradictory). *Let $P_{q\bar{r}} \in \text{PrePol}_{q\bar{r}}^{\text{res}}$ be a package whose ordinary readout claims an ordinary top-bearing meson, and let $\mathcal{R}_{\text{ch}} \in \text{ChResType}$. If $\text{TopCh}(q\bar{r})$ holds and $\text{WCl}(q\bar{r}) \not\subseteq \mathcal{R}_{\text{ch}}(q\bar{r})$, then the ordinary-meson readout symbol is contradictory. Equivalently,*

$$\text{TopCh}(q\bar{r}) \wedge \text{WCl}(q\bar{r}) \not\subseteq \mathcal{R}_{\text{ch}}(q\bar{r}) \implies \text{Contr}_{\text{top}}(P_{q\bar{r}}; \mathcal{R}_{\text{ch}}).$$

Definition 44 (Grand Meson charge-closure vector). In the Grand Meson calculus, “charge” means a charge-closure vector, not electric charge alone. Fix a finite sector-label space $\mathbb{R}^{\Sigma_{\text{GM}}}$ containing the spin-parity, threshold, anomaly, and readout labels used in the paper, and set

$$\mathcal{K}_{\text{GM}} := \mathbb{R} \oplus \mathbb{R}^{F_6} \oplus \mathbb{R}^{3 \times 3} \oplus \mathbb{R} \oplus \mathbb{R}^{\Sigma_{\text{GM}}}.$$

Equip this finite-dimensional real vector space with a fixed weighted inner product $\langle -, - \rangle_W$. The components below are embedded into the displayed direct summands. For any pair

$$\Xi = (\sigma^\Xi, \mathcal{R}_{\text{ch}}^\Xi)$$

with

$$\sigma^\Xi = (\sigma_{q\bar{r}}^\Xi)_{q, r \in F_6}, \quad \mathcal{R}_{\text{ch}}^\Xi \in \text{ChResType},$$

define

$$\tau_{\text{top}}^\Xi(q, \bar{r}) := \begin{cases} 1, & \text{TopCh}(q\bar{r}) \text{ and } \text{WCl}(q\bar{r}) \not\subseteq \mathcal{R}_{\text{ch}}^\Xi(q\bar{r}), \\ 0, & \text{otherwise.} \end{cases}$$

For $q \in A_i$, $r \in A_j$, and spin-parity sector σ , define

$$\kappa_{\text{GM}}^\Xi(q, \bar{r}; \sigma) := (Q_q - Q_r, e_q - e_r, \mathbf{a}_{ij}, \tau_{\text{top}}^\Xi(q, \bar{r}), \chi_\sigma(q, \bar{r})) \in \mathcal{K}_{\text{GM}}.$$

For an oriented channel $c = (q, \bar{r})$, write

$$\kappa_{\text{GM}}^\Xi(c) := \kappa_{\text{GM}}^\Xi(q, \bar{r}; \sigma_{q\bar{r}}^\Xi).$$

The coordinate $\chi_\sigma(q, \bar{r})$ records the spin-parity, threshold, anomaly, and readout sector used by the package. For each channel and sector let

$$\mathcal{Z}_{q\bar{r}, \sigma} \subset \mathcal{K}_{\text{GM}}$$

be a nonempty closed affine subspace, called the closed charge-balanced readout locus, and let

$$\Pi_{q\bar{r},\sigma} : \mathcal{K}_{\text{GM}} \rightarrow \mathcal{Z}_{q\bar{r},\sigma}$$

be the unique W -orthogonal projection onto it.

Definition 45 (Charge-closure strain and stability). The following structure is an added diagnostic layer on top of the channel-index lift. It is not derived from QCD dynamics or from the existence of PreBody_{36} . It supplies a closure-distance semantics for robustness tests inside this paper.

Let $G \in \text{PreBody}_{36}$ be an inspectable Grand Meson body. A Grand Meson charge assignment is a variable family

$$\mathbf{K} = (K_{q\bar{r}})_{q,r \in F_6}, \quad K_{q\bar{r}} \in \mathcal{K}_{\text{GM}},$$

and the diagnostic configuration supplies the channelwise sector and channel residual-closure assignments:

$$\Xi := (\boldsymbol{\sigma}^\Xi, \mathcal{R}_{\text{ch}}^\Xi), \quad \boldsymbol{\sigma}^\Xi = (\sigma_{q\bar{r}}^\Xi)_{q,r \in F_6},$$

where $\mathcal{R}_{\text{ch}}^\Xi$ is a fixed channel residual-closure assignment with $\mathcal{R}_{\text{ch}}^\Xi \in \text{ChResType}$. The distinguished standard assignment for Ξ is

$$K_{q\bar{r}}^{\text{std},\Xi} := \kappa_{\text{GM}}^\Xi(q, \bar{r}; \sigma_{q\bar{r}}^\Xi).$$

All strain, projection, and stability expressions in this section carry Ξ explicitly. Define channel activity by

$$a_G(q, \bar{r}) := \begin{cases} 0, & G(q, \bar{r}) = \mathbf{0}_{q\bar{r}}[\rho] \text{ for some } \rho \in \text{Prov}_0, \\ 1, & \text{otherwise.} \end{cases}$$

Define

$$\text{TopOk}_\Xi(q\bar{r}) \iff \neg \text{TopCh}(q\bar{r}) \text{ or } \text{WCl}(q\bar{r}) \subseteq \mathcal{R}_{\text{ch}}^\Xi(q\bar{r}).$$

The closure barrier is

$$\Theta_{G,\Xi}(q, \bar{r}) := \begin{cases} 0, & a_G(q, \bar{r}) = 0, \\ 0, & a_G(q, \bar{r}) = 1 \text{ and } \text{Cl}(G(q, \bar{r})) = \text{Cl}_{\text{GM}} \text{ and } \text{TopOk}_\Xi(q\bar{r}), \\ +\infty, & \text{otherwise.} \end{cases}$$

The charge-closure imbalance of $q\bar{r}$, valued in the extended nonnegative reals, is

$$\mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K}) := a_G(q, \bar{r}) \text{dist}_W^2(K_{q\bar{r}}, \mathcal{Z}_{q\bar{r}, \sigma_{q\bar{r}}^\Xi}) + \Theta_{G,\Xi}(q, \bar{r}),$$

where

$$\text{dist}_W^2(K_{q\bar{r}}, \mathcal{Z}_{q\bar{r}, \sigma_{q\bar{r}}^\Xi}) := \left\| K_{q\bar{r}} - \Pi_{q\bar{r}, \sigma_{q\bar{r}}^\Xi} K_{q\bar{r}} \right\|_W^2.$$

Write $\varepsilon_{G,\Xi}(q, \bar{r}; \mathbf{K}) := \mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K})$. For an oriented channel $c = (q, \bar{r})$, write

$$\mathcal{I}_{G,\Xi}(c; \mathbf{K}) := \mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K}).$$

The total Grand Meson strain and stability are

$$\mathcal{E}_{G,\Xi}(\mathbf{K}) \in [0, +\infty] := \sum_{q,r \in F_6} \mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K}), \quad \text{St}_{G,\Xi}(\mathbf{K}) := \frac{1}{1 + \mathcal{E}_{G,\Xi}(\mathbf{K})}.$$

Thus $\text{St}_{G,\Xi}(\mathbf{K}) = 0$ when an unclosed channel carries the $+\infty$ barrier. The imbalance is a charge-closure imbalance: electric charge, flavor, axis-pair data, top obstruction, and the readout sector all participate.

Definition 46 (Transition margin). For an oriented channel $q\bar{r}$, define the finite candidate target set

$$\text{Cand}_{G,\Xi}(q\bar{r}; \mathbf{K}) := \{ s\bar{t} \mid \mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K}) < +\infty, \mathcal{I}_{G,\Xi}(s, \bar{t}; \mathbf{K}) < +\infty \}.$$

For $s\bar{t} \in \text{Cand}_{G,\Xi}(q\bar{r}; \mathbf{K})$, define the Grand Meson transition margin

$$M_{G,\Xi}(q\bar{r} \rightarrow s\bar{t}; \mathbf{K}) := \mathcal{I}_{G,\Xi}(s, \bar{t}; \mathbf{K}) - \mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K}).$$

Among closed candidate targets, the internally preferred directions in this diagnostic order are the minimizers

$$\arg \min_{s\bar{t} \in \text{Cand}_{G,\Xi}(q\bar{r}; \mathbf{K})} M_{G,\Xi}(q\bar{r} \rightarrow s\bar{t}; \mathbf{K}).$$

This is an internal preference order, not a decay-rate prediction.

Definition 47 (Projected discontinuity certificate). On the product space

$$\prod_{q,r \in F_6} \mathcal{K}_{\text{GM}},$$

use the product weighted inner product

$$\langle H, H' \rangle_{\Pi W} := \sum_{q,r \in F_6} \langle H_{q\bar{r}}, H'_{q\bar{r}} \rangle_W.$$

Let $\text{Sm}_{G,\Xi}$ be the finite smooth projection stratum on which $\mathcal{E}_{G,\Xi}(\mathbf{K}) < +\infty$ and every projection $\Pi_{q\bar{r}, \sigma_{q\bar{r}}^\Xi}$ appearing in an active channel is differentiable. The UD-visible strain is

$$\mathcal{E}_{Y,G,\Xi}(\mathbf{K}) := \sum_{q,r \in A_Y} \mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K}),$$

and the non-UD residual strain opened by the Grand Meson body is

$$\mathcal{E}_{\perp,G,\Xi}(\mathbf{K}) := \mathcal{E}_{G,\Xi}(\mathbf{K}) - \mathcal{E}_{Y,G,\Xi}(\mathbf{K}) = \sum_{(q,\bar{r}) \notin A_Y \times \overline{A_Y}} \mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K}).$$

Define the static discontinuity certificate

$$\mathfrak{D}_{G,\Xi} : \text{Sm}_{G,\Xi} \rightarrow [0, +\infty) \times \prod_{q,r \in F_6} \mathcal{K}_{\text{GM}}, \quad \mathfrak{D}_{G,\Xi}(\mathbf{K}) := (\mathcal{E}_{\perp,G,\Xi}(\mathbf{K}), \nabla \mathcal{E}_{\perp,G,\Xi}(\mathbf{K})).$$

It measures not a dangerous object, but a projected network imbalance: the part of the charge-closure network that is invisible before the non-UD axes are opened.

Definition 48 (Fuse set and fuse normal form). Fix a readout threshold $\epsilon_{\text{fuse}} > 0$. After the non-UD axes are opened, define the fuse set

$$\Omega_{\text{fuse}}(G, \Xi, \mathbf{K}) := \{ (q, \bar{r}) \mid \mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K}) = +\infty \text{ or } \mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K}) \geq \epsilon_{\text{fuse}} \}.$$

The non-UD residual fuse set is

$$\Omega_{\perp}(G, \Xi, \mathbf{K}) := \Omega_{\text{fuse}}(G, \Xi, \mathbf{K}) \cap ((F_6 \times \overline{F_6}) \setminus (A_Y \times \overline{A_Y})).$$

The associated residual fission network is the directed graph

$$E_{\perp}(G, \Xi, \mathbf{K}) := \{ (q, r) \in F_6 \times F_6 \mid (q, \bar{r}) \in \Omega_{\perp}(G, \Xi, \mathbf{K}) \}.$$

$$\mathcal{N}_\perp(G, \Xi, \mathbf{K}) := (F_6, E_\perp(G, \Xi, \mathbf{K})).$$

The fuse normal form is the body map

$$\text{Fuse}_{G, \Xi, \mathbf{K}} \in \text{Body}_{36}^{\text{cl}} \quad \text{defined by} \quad \text{Fuse}_{G, \Xi, \mathbf{K}}(q, \bar{r}) := \begin{cases} \mathbf{0}_{q\bar{r}}[\text{fuse}], & (q, \bar{r}) \in \Omega_{\text{fuse}}(G, \Xi, \mathbf{K}), \\ G(q, \bar{r}), & \text{otherwise.} \end{cases}$$

This is not a destructive physical process. It is the Grand Meson closure response that turns a detected charge-closure imbalance into a closed inactive residual package. The codomain is closed: every fuse edge is replaced by $\mathbf{0}[\text{fuse}]$, while every non-fuse active edge has finite barrier and therefore satisfies the Grand closure suite.

Proposition 49 (Closedness of the fuse normal form). *For every $G \in \text{PreBody}_{36}$, diagnostic configuration Ξ , and $\mathbf{K}$, the body map $\text{Fuse}_{G, \Xi, \mathbf{K}}$ lies in $\text{Body}_{36}^{\text{cl}}$.*

Proof. If $(q, \bar{r}) \in \Omega_{\text{fuse}}(G, \Xi, \mathbf{K})$, the output component is $\mathbf{0}_{q\bar{r}}[\text{fuse}]$, hence a closed inactive residual package. If $(q, \bar{r}) \notin \Omega_{\text{fuse}}(G, \Xi, \mathbf{K})$, then $\mathcal{I}_{G, \Xi}(q, \bar{r}; \mathbf{K}) < +\infty$. Since the only infinite contribution to $\mathcal{I}_{G, \Xi}$ is the closure barrier, $\Theta_{G, \Xi}(q, \bar{r}) = 0$. Therefore the component is either inactive, or it is active and satisfies $\text{Cl} = \text{Cl}_{\text{GM}}$ together with TopOk_Ξ . Thus every output component is closed. $\square$

Definition 50 (Finite readout load). Let $H \in \text{PreBody}_{36}$ be any proposed post-opening body map that agrees with G outside $\Omega_{\text{fuse}}(G, \Xi, \mathbf{K})$. Define the residual readout load on an imbalanced edge by

$$\lambda_{G, \Xi, \mathbf{K}}^H(q, \bar{r}) := \begin{cases} 0, & H(q, \bar{r}) = \mathbf{0}_{q\bar{r}}[\text{fuse}], \\ +\infty, & H(q, \bar{r}) \neq \mathbf{0}_{q\bar{r}}[\text{fuse}], \end{cases} \quad (q, \bar{r}) \in \Omega_{\text{fuse}}(G, \Xi, \mathbf{K}).$$

The total residual readout load is

$$\text{Load}_{G, \Xi, \mathbf{K}}(H) := \sum_{(q, \bar{r}) \in \Omega_{\text{fuse}}(G, \Xi, \mathbf{K})} \lambda_{G, \Xi, \mathbf{K}}^H(q, \bar{r}).$$

A finite Grand Meson readout circuit is one whose residual readout load is finite. For a finite cutoff $L > 0$, define

$$\lambda_{G, \Xi, \mathbf{K}}^{H, L}(q, \bar{r}) := \begin{cases} 0, & H(q, \bar{r}) = \mathbf{0}_{q\bar{r}}[\text{fuse}], \\ L, & H(q, \bar{r}) \neq \mathbf{0}_{q\bar{r}}[\text{fuse}], \end{cases}$$

and

$$\text{Load}_{G, \Xi, \mathbf{K}}^{(L)}(H) := \sum_{(q, \bar{r}) \in \Omega_{\text{fuse}}(G, \Xi, \mathbf{K})} \lambda_{G, \Xi, \mathbf{K}}^{H, L}(q, \bar{r}).$$

Definition 51 (Body transition space). A body transition symbol is written $U : G \rightarrow G'$, with $G, G' \in \text{PreBody}_{36}$. Its changed body support is

$$\Delta_{\text{body}}(U) := \{ (q, \bar{r}) \mid G(q, \bar{r}) \neq G'(q, \bar{r}) \}.$$

After a fuse network has been opened, the support that must be closed is not only the changed support but also the visible fuse support:

$$\Sigma_{\text{body}}(U; G, \Xi, \mathbf{K}) := \Delta_{\text{body}}(U) \cup \Omega_{\text{fuse}}(G, \Xi, \mathbf{K}).$$

The body transition space is

$$\text{Mor}_{\text{GM}}^{\text{body}}(G, \Xi, \mathbf{K}) := \{ U : G \rightarrow G' \mid \Sigma_{\text{body}}(U; G, \Xi, \mathbf{K}) \subseteq \text{RCl}_{\text{body}}(U) \}.$$

The body-transition contradiction predicate is

$$\text{Contr}_{\text{tr}}(U; G, \Xi, \mathbf{K}) := \Sigma_{\text{body}}(U; G, \Xi, \mathbf{K}) \setminus \text{RCl}_{\text{body}}(U) \neq \emptyset.$$

Theorem 52 (Deterministic charge-closure strain diagnostic). *Fix an inspectable Grand Meson body G , a diagnostic configuration Ξ , and a Grand Meson charge assignment $\mathbf{K} \in \mathbf{Sm}_{G,\Xi}$. Then the following data are determined uniquely:*

$$\mathbf{St}_{G,\Xi}(\mathbf{K}), \quad \mathfrak{D}_{G,\Xi}(\mathbf{K}), \quad M_{G,\Xi}(q\bar{r} \rightarrow s\bar{t}; \mathbf{K}),$$

for every finite candidate $s\bar{t} \in \mathbf{Cand}_{G,\Xi}(q\bar{r}; \mathbf{K})$. On every finite smooth projection stratum, for a variation $h = (h_{q\bar{r}})_{q,r \in F_6}$,

$$D\mathcal{E}_{G,\Xi}(\mathbf{K})[h] = 2 \sum_{q,r \in F_6} a_G(q, \bar{r}) \left\langle K_{q\bar{r}} - \Pi_{q\bar{r}, \sigma_{q\bar{r}}^\Xi} K_{q\bar{r}}, h_{q\bar{r}} - D\Pi_{q\bar{r}, \sigma_{q\bar{r}}^\Xi}(K_{q\bar{r}})[h_{q\bar{r}}] \right\rangle_W.$$

Consequently

$$D\mathbf{St}_{G,\Xi}(\mathbf{K})[h] = -\mathbf{St}_{G,\Xi}(\mathbf{K})^2 D\mathcal{E}_{G,\Xi}(\mathbf{K})[h].$$

If the channel charges are functions of a common parameter θ , namely $\mathbf{K} = \mathbf{K}(\theta)$, then

$$\frac{d}{d\theta} \mathcal{E}_{G,\Xi}(\mathbf{K}(\theta)) = D\mathcal{E}_{G,\Xi}(\mathbf{K}(\theta)) \left[\frac{d\mathbf{K}}{d\theta} \right].$$

Thus a change in one physical parameter propagates deterministically through exactly the channels whose charge-closure vectors depend on that parameter. No extra prohibition rule is required: an unclosed top-bearing or otherwise unclosed active channel receives the $+\infty$ barrier and therefore cannot inhabit a stable Grand Meson transition.

Finally, the UD projection detects exactly the UD-visible part and no more: for any $\mathbf{K}, \mathbf{K}'$ with

$$K_{q\bar{r}} = K'_{q\bar{r}} \quad (q, r \in A_Y),$$

one has

$$\mathcal{E}_{Y,G,\Xi}(\mathbf{K}) = \mathcal{E}_{Y,G,\Xi}(\mathbf{K}'), \quad D\mathcal{E}_{Y,G,\Xi}(\mathbf{K}) = D\mathcal{E}_{Y,G,\Xi}(\mathbf{K}') \text{ on } A_Y \times \overline{A_Y}.$$

Any change in

$$\mathcal{E}_{G,\Xi} - \mathcal{E}_{Y,G,\Xi} = \mathcal{E}_{\perp,G,\Xi}$$

is therefore statically invisible to the General Meson and becomes verifiable only after the non-UD axes are opened. Along any differentiable path $\gamma : [0, 1] \rightarrow \mathbf{Sm}_{G,\Xi}$,

$$\int_{\gamma} D\mathcal{E}_{G,\Xi} - \int_{\gamma} D\mathcal{E}_{Y,G,\Xi} = \mathcal{E}_{\perp,G,\Xi}(\gamma(1)) - \mathcal{E}_{\perp,G,\Xi}(\gamma(0)).$$

Thus a projected differential or integral mismatch is exactly the non-UD residual certificate $\mathfrak{D}_{G,\Xi}$. Instability in this calculus is not assigned by a name; it is the nonzero, threshold-level, or infinite charge-closure imbalance carried by typed mesonic channels.

Proof. The activity a_G , the projection data $\Pi_{q\bar{r}, \sigma_{q\bar{r}}^\Xi}$, the balanced loci $\mathcal{Z}_{q\bar{r}, \sigma_{q\bar{r}}^\Xi}$, and the barrier $\Theta_{G,\Xi}$ are functions of the fixed data $(G, \Xi, \mathbf{K})$. Hence every channel imbalance $\mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K})$, their sum $\mathcal{E}_{G,\Xi}(\mathbf{K})$, the UD-visible sum $\mathcal{E}_{Y,G,\Xi}(\mathbf{K})$, the residual sum $\mathcal{E}_{\perp,G,\Xi}(\mathbf{K})$, the certificate $\mathfrak{D}_{G,\Xi}(\mathbf{K})$, the stability $\mathbf{St}_{G,\Xi}(\mathbf{K})$, and every transition margin $M_{G,\Xi}$ are uniquely determined.

On a finite smooth projection stratum,

$$\mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K}) = a_G(q, \bar{r}) \|K_{q\bar{r}} - \Pi_{q\bar{r}, \sigma_{q\bar{r}}^\Xi} K_{q\bar{r}}\|_W^2.$$

Differentiating this weighted square gives the displayed formula for $D\mathcal{E}_{G,\Xi}$. Since $\mathbf{St}_{G,\Xi} = (1 + \mathcal{E}_{G,\Xi})^{-1}$, the chain rule gives the displayed formula for $D\mathbf{St}_{G,\Xi}$. If a required closure is

missing, $\Theta_{G,\Xi} = +\infty$; then the channel is not a finite stable transition candidate inside the Grand Meson calculus.

The UD-visible strain $\mathcal{E}_{Y,G,\Xi}$ is a sum over $A_Y \times \overline{A_Y}$, so it depends only on the UD restriction of $\mathbf{K}$. Therefore two assignments with the same UD restriction have the same $\mathcal{E}_{Y,G,\Xi}$ and the same UD differential. Subtracting $\mathcal{E}_{Y,G,\Xi}$ from $\mathcal{E}_{G,\Xi}$ leaves exactly the sum of all non-UD channel imbalances, which is $\mathcal{E}_{\perp,G,\Xi}$. Exactness of the differential of the scalar strain along γ gives the displayed integral mismatch. Thus the opened Grand Meson axes do not add a rhetorical warning; they add a static certificate for the residual imbalance that the UD projection cannot see. $\square$

10 Finite Fuse Normalization

Finite fuse normalization is the finite-readout form of the Grand Meson closure discipline. Once a charge-closure fission network is made visible by opening the non-UD axes, the network cannot remain as active readout inside a finite Grand Meson circuit. It automatically normalizes to closed fuse residual packages; if one insists on keeping it active, the readout load becomes non-finite.

Theorem 53 (Finite fuse normal form). *Assume all channels have been safely opened, meaning that every $(q, \bar{r}) \in F_6 \times \overline{F_6}$ is typed either by an active package in $\text{Pol}_{q\bar{r}}$ or by an inactive package $\mathbf{0}_{q\bar{r}}[\rho] \in \text{Pol}_{q\bar{r}}^{\text{res}}$. If*

$$\Omega_{\text{fuse}}(G, \Xi, \mathbf{K}) \neq \emptyset,$$

then the finite Grand Meson readout circuit has a unique normal form on the opened imbalanced network:

$$\text{Load}_{G,\Xi,\mathbf{K}}(H) < +\infty \iff H(q, \bar{r}) = \text{Fuse}_{G,\Xi,\mathbf{K}}(q, \bar{r}) \text{ for all } (q, \bar{r}) \in \Omega_{\text{fuse}}(G, \Xi, \mathbf{K}).$$

Thus the channel does not decide to cut itself; within the finite-load semantics defined here, any non-fuse value has infinite residual readout load.

Equivalently, no Grand Meson body transition may ignore the opened imbalance. Let $U : G \rightarrow G'$ preserve the UD body projection and leave at least one imbalanced edge unclosed:

$$p_{\text{ud}}^{\text{body}}(G') = p_{\text{ud}}^{\text{body}}(G), \quad \Omega_{\text{fuse}}(G, \Xi, \mathbf{K}) \setminus \text{RCl}_{\text{body}}(U) \neq \emptyset.$$

Then

$$U \notin \text{Mor}_{\text{GM}}^{\text{body}}(G, \Xi, \mathbf{K}), \quad \text{Contr}_{\text{tr}}(U; G, \Xi, \mathbf{K}).$$

Moreover, $\text{Fuse}_{G,\Xi,\mathbf{K}}$ deterministically closes every opened fuse edge:

$$(q, \bar{r}) \in \Omega_{\text{fuse}}(G, \Xi, \mathbf{K}) \implies \text{Fuse}_{G,\Xi,\mathbf{K}}(q, \bar{r}) = \mathbf{0}_{q\bar{r}}[\text{fuse}].$$

In a finite cutoff semantics where $+\infty$ is replaced by L , any body map H leaving k fuse edges unfused has

$$\text{Load}_{G,\Xi,\mathbf{K}}^{(L)}(H) \geq kL.$$

Hence the computational or cognitive burden of retaining the large-scale fission network as active readout cannot remain bounded as $L \rightarrow \infty$. Once the opened charge-closure imbalance is visible, the finite readout circuit automatically collapses to the closed fuse residual form.

Proof. Safe opening places every oriented channel in the typed body $\text{PrePol}_{q\bar{r}}^{\text{res}}$, so every fuse edge in $\Omega_{\text{fuse}}(G, \Xi, \mathbf{K})$ is a readable element of the Grand Meson body rather than a missing symbol. By definition of the residual readout load, each fuse edge contributes 0 exactly when its post-opening value is $\mathbf{0}_{q\bar{r}}[\text{fuse}]$, and contributes $+\infty$ otherwise. Therefore a proposed body

map has finite load exactly when it agrees with $\text{Fuse}_{G,\Xi,\mathbf{K}}$ on every edge of $\Omega_{\text{fuse}}(G, \Xi, \mathbf{K})$. This proves uniqueness of the finite normal form.

If a transition U preserves the UD body projection while leaving some $(q, \bar{r}) \in \Omega_{\text{fuse}}(G, \Xi, \mathbf{K})$ outside $\text{RCl}_{\text{body}}(U)$, then the changed or retained residual support is not contained in the residual closure set required for membership in $\text{Mor}_{\text{GM}}^{\text{body}}(G, \Xi, \mathbf{K})$. By the definition of the opened body transition space, $U \notin \text{Mor}_{\text{GM}}^{\text{body}}(G, \Xi, \mathbf{K})$, and by the definition of the contradiction predicate the same unclosed edge witnesses $\text{Contr}_{\text{tr}}(U; G, \Xi, \mathbf{K})$.

The displayed implication for $\text{Fuse}_{G,\Xi,\mathbf{K}}$ is its definition. For every edge in Ω_{fuse} , the output is $\mathbf{0}_{q\bar{r}}[\text{fuse}]$, which belongs to $\text{Pol}_{q\bar{r}}^{\text{res}}$ and satisfies $\text{Cl}(\mathbf{0}_{q\bar{r}}[\text{fuse}]) = \text{Cl}_{\text{GM}}$. For every edge outside Ω_{fuse} , the channel is left unchanged. Hence the operation closes every detected fuse edge, while an already fuse-tagged inactive input outside Ω_{fuse} is not used as evidence that the edge belongs to the fuse set. Under a finite cutoff L , each of the k unfused fuse edges contributes at least L to the load, giving the bound $\text{Load}_{G,\Xi,\mathbf{K}}^{(L)}(H) \geq kL$. The load is therefore unbounded in the limit $L \rightarrow \infty$. $\square$

Definition 54 (Grand Meson decay readout). For a finite final readout family $\mathcal{F} = \{((q_y, \bar{r}_y), \eta_y)\}$, where each $(q_y, \bar{r}_y)$ is a typed oriented output channel and $\eta_y \in \{\pm 1\}$, write

$$\kappa_{\text{GM}}^{\Xi}(\mathcal{F}) := \sum_{((q_y, \bar{r}_y), \eta_y) \in \mathcal{F}} \eta_y \kappa_{\text{GM}}^{\Xi}(q_y, \bar{r}_y; \sigma_{q_y \bar{r}_y}^{\Xi}), \quad \mathcal{I}_{G,\Xi}(\mathcal{F}; \mathbf{K}) := \sum_{((q_y, \bar{r}_y), \eta_y) \in \mathcal{F}} \mathcal{I}_{G,\Xi}(q_y, \bar{r}_y; \mathbf{K}).$$

A Grand Meson decay readout of an active channel $c = (q, \bar{r})$ into $\mathcal{F}_c$ is a closed readout family satisfying

$$\kappa_{\text{GM}}^{\Xi}(c) = \kappa_{\text{GM}}^{\Xi}(\mathcal{F}_c) + R_{\text{closed}}^{\kappa}, \quad \mathcal{I}_{G,\Xi}(\mathcal{F}_c; \mathbf{K}) < \mathcal{I}_{G,\Xi}(c; \mathbf{K}),$$

where $R_{\text{closed}}^{\kappa} \in \mathcal{K}_{\text{GM}}$ is the charge-closure vector of a closed residual record. For a channel c , write

$$R_c^{\text{fuse}} := \mathbf{0}_c[\text{fuse}]$$

for the fuse residual record of c .

Theorem 55 (Fuse residual attached to a declared decay readout). *Let $c = (q, \bar{r})$ be an active mesonic channel with $c \in \Omega_{\text{fuse}}(G, \Xi, \mathbf{K})$. If there exists a Grand Meson decay readout family $\mathcal{F}_c$, then every finite Grand Meson readout H of c has the fuse residual record as its channel value:*

$$H(c) = R_c^{\text{fuse}}.$$

In the present calculus, the declared decay readout is represented not as deletion but as a fuse residual record attached to a closed readout family. The family $\mathcal{F}_c$ remains part of the separately declared closed readout relation above.

Proof. Since $c \in \Omega_{\text{fuse}}(G, \Xi, \mathbf{K})$, the finite fuse normal form says that any finite readout must agree with the fuse normal form on c . Hence $H(c) = \mathbf{0}_c[\text{fuse}] = R_c^{\text{fuse}}$. The assumed family $\mathcal{F}_c$ is not derived from the zero object; it is an additional closed readout relation whose charge vector is conserved up to $R_{\text{closed}}^{\kappa}$ and whose imbalance is strictly lower. $\square$

Conjecture 56 (Possible diagnostic comparison with physical decay widths). For ordinary non-top mesonic channels for which a Standard Model decay amplitude exists, the Grand Meson imbalance $\mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K})$ is a closure-index shadow of the physical decay width. This is a proposed external calibration problem, not a result of the Grand Meson calculus. More precisely, after fixing the spin-parity sector and available final states, there is a monotone readout functional $\mathcal{R}$ such that

$$\Gamma_{\text{phys}}(q\bar{r} \rightarrow f) = \mathcal{R}(\mathcal{I}_{G,\Xi}(q, \bar{r}; \mathbf{K}), \rho_f, |\mathcal{A}_{\text{SM}}(q\bar{r} \rightarrow f)|^2),$$

where ρ_f is the final-state phase space and $\mathcal{A}_{\text{SM}}$ is the corresponding Standard Model amplitude. This conjecture does not replace the Standard Model decay formula; it asks whether the Grand Meson fuse index can be read as a closure layer above the ordinary amplitude.

Definition 57 (Anti-face fission). Let the opposite face on the same generation axis be

$$\text{op}(\Phi(i, \epsilon)) := \Phi(i, -\epsilon), \quad i \in G_3, \epsilon \in H_2.$$

For a mesonic channel $O_{q\bar{r}}$, the anti-quark side is the charge conjugate $\bar{r}$. It is not the opposite Grand Meson face $\overline{\text{op}(r)}$. Define the anti-face fission candidate by

$$\text{Fis}_{\bar{F}}(O_{q\bar{r}}) := O_{q\overline{\text{op}(r)}}.$$

This symbol is not a new physical process. It is the diagnostic pattern in which the anti-quark side of a channel is replaced by the opposite Grand Meson face on the same generation axis.

Axiom 58 (Sector-respecting identity). Grand Meson identity preserves the charge–flavor fingerprint:

$$O_{q\bar{r}} \equiv_{\text{GM}} O_{s\bar{t}} \implies \nu_{\text{QCD}}(q, \bar{r}; \sigma) = \nu_{\text{QCD}}(s, \bar{t}; \sigma).$$

Thus two distinct quantum-number sectors may be related by an explicit transition with closed residue, but they may not be silently identified.

Proposition 59 (Anti-face fission is a Grand Meson contradiction). *Let*

$$\kappa(q, \bar{r}) := (Q_q - Q_r, e_q - e_r)$$

be the charge–flavor fingerprint of the mesonic channel, where Q_q is the electric charge of q and e_q is its flavor unit. If

$$R_{\bar{F}}(q, \bar{r}) := \kappa(q, \bar{r}) - \kappa(q, \overline{\text{op}(r)}) \neq 0$$

then identifying $O_{q\bar{r}}$ with $\text{Fis}_{\bar{F}}(O_{q\bar{r}})$ is contradictory:

$$O_{q\bar{r}} \equiv_{\text{GM}} \text{Fis}_{\bar{F}}(O_{q\bar{r}}) \implies \text{Contr}_{\bar{F}}(O_{q\bar{r}} \equiv_{\text{GM}} \text{Fis}_{\bar{F}}(O_{q\bar{r}})).$$

Proof. The Grand Meson readout preserves the ordered pair consisting of the quark side and the anti-quark side. Replacing $\bar{r}$ by $\overline{\text{op}(r)}$ changes the channel fingerprint by $R_{\bar{F}}(q, \bar{r})$. When this residual fingerprint is nonzero, the two channels lie in distinct charge–flavor sectors, so the proposed identification cannot inhabit the Grand Meson readout space. $\square$

Remark 60 (Examples). For the light pion channel, anti-face fission sends

$$O_{u\bar{d}} \mapsto O_{u\bar{u}}.$$

The electric charge changes from $Q_u - Q_d = +1$ to $Q_u - Q_u = 0$, so the residual fingerprint is nonzero. For a cross-generation channel,

$$O_{u\bar{s}} \mapsto O_{u\bar{c}},$$

the same block occurs: $Q_u - Q_s = +1$, whereas $Q_u - Q_c = 0$. On the third axis,

$$O_{u\bar{b}} \mapsto O_{u\bar{t}},$$

the flipped readout also enters the top-axis closure condition above. Thus the Grand Meson is not merely an untyped relabeling; it adds typed closure tests that expose wrong anti-face identifications. The flipped symbol may only be treated as a distinct channel with its own package and quantum fingerprint.

Remark 61 (Worked channel table). The following examples show that the completion is operational rather than merely decorative. “Ord.” means ordinary hadronic meson readout, and $R_\perp$ is the residual axis content under downgrade to the $u \leftrightarrow d$ face.

channel	$q\bar{r}$	axes	ord.	top	anti-face test / $R_\perp$
π^+	$u\bar{d}$	(1, 1)	yes	no	$u\bar{u} : Q + 1 \rightarrow 0; R_\perp = 0$
K^+	$u\bar{s}$	(1, 2)	yes	no	$u\bar{c} : Q + 1 \rightarrow 0; R_\perp = A_2$
D^0	$c\bar{u}$	(2, 1)	yes	no	$c\bar{d} : Q0 \rightarrow +1; R_\perp = A_2$
D_s^+	$c\bar{s}$	(2, 2)	yes	no	$c\bar{c} : Q + 1 \rightarrow 0; R_\perp = A_2$
B^-	$b\bar{u}$	(3, 1)	yes	no	$b\bar{d} : Q - 1 \rightarrow 0; R_\perp = A_3$
J/ψ	$c\bar{c}$	(2, 2)	yes	no	$c\bar{s} : Q0 \rightarrow +1; R_\perp = A_2$
Υ	$b\bar{b}$	(3, 3)	yes	no	$b\bar{t} : Q0 \rightarrow -1; R_\perp = A_3$
$t\bar{b}$	$t\bar{b}$	(3, 3)	no	WCI	$t\bar{t} : Q + 1 \rightarrow 0; R_\perp = A_3$
$u\bar{t}$	$u\bar{t}$	(1, 3)	no	WCI	$u\bar{b} : Q0 \rightarrow +1; R_\perp = A_3$

The first seven rows are ordinary QCD mesonic readouts after closure. The last two are not ordinary hadronic mesons in this framework; they are top-bearing weak-virtual channels. In every row, the anti-face candidate changes the charge-flavor sector, so an attempted identification is a $\text{Contr}_{\bar{F}}$ witness unless the residual channel is explicitly closed by a distinct transition.

Definition 62 (Grand Meson CKM lift). Let

$$U_L = (u_L, c_L, t_L)^T, \quad D_L = (d_L, s_L, b_L)^T,$$

and let $V_{\text{CKM}} = (V_{ij}) \in U(3)$ be the charged-current flavor-mixing matrix. The ordinary weak current is

$$J_+^\mu = \bar{U}_L \gamma^\mu V_{\text{CKM}} D_L = \sum_{i,j=1}^3 V_{ij} \bar{u}_{iL} \gamma^\mu d_{jL}.$$

The Grand Meson lift places each CKM coefficient on its axis pair:

$$\Gamma_{\text{GM}}^{\text{CKM}}(V_{ij}) := (i, j, V_{ij}).$$

For a channel package $P_{q\bar{r}}$, the CKM closure predicate $\text{CKMCl}(P_{q\bar{r}})$ is not evaluated on the meson axis pair $(q, \bar{r})$ itself. It is evaluated on each weak charged-current edge actually used inside the package. If such an edge has the form $\bar{u}_{iL} \gamma^\mu d_{jL}$, then that edge must be annotated by (i, j, V_{ij}) and must recover V_{ij} after forgetting the Grand Meson labels. If the package uses no weak charged-current edge, the CKM predicate is vacuous.

Proposition 63 (CKM recovery). *Let $\pi_{\text{CKM}}(i, j, V_{ij}) = V_{ij}$. Then forgetting the Grand Meson axis labels recovers the ordinary CKM current:*

$$\sum_{i,j} \pi_{\text{CKM}}(\Gamma_{\text{GM}}^{\text{CKM}}(V_{ij})) \bar{u}_{iL} \gamma^\mu d_{jL} = J_+^\mu.$$

Thus the Grand Meson does not derive the CKM matrix. It annotates the charged-current terms by generation-axis labels and verifies that forgetting those labels recovers the ordinary weak current. CKM mixing is therefore a required weak-current closure layer, not a modification of QCD color dynamics.

The ordinary QCD mesonic operator basis is

$$\mathbf{Q}_{\text{mes}} := \left\{ O_{q\bar{r}}^{\Gamma_\sigma}(x) = \sum_{a=1}^3 \bar{r}_a(x) \Gamma_\sigma q^a(x) \mid q, r \in F_6, \sigma \text{ a spin-parity sector} \right\}.$$

Standard channels are recovered inside this basis, for example

$$\pi^+ \sim O_{u\bar{d}}, \quad K^+ \sim O_{u\bar{s}}, \quad D^0 \sim O_{c\bar{u}}, \quad B^- \sim O_{b\bar{u}},$$

and

$$J/\psi \sim O_{c\bar{c}}, \quad \Upsilon \sim O_{b\bar{b}}, \quad \eta, \eta' \sim \text{span}\{O_{u\bar{u}}, O_{d\bar{d}}, O_{s\bar{s}}\}.$$

Thus cross-generation mesons such as $u\bar{s}$, $c\bar{d}$, $b\bar{u}$, and $c\bar{s}$ are ordinary members of the same basis. The Grand Meson lift is

$$\Gamma_{\text{GM}}(O_{q\bar{r}}) := (i, j, O_{q\bar{r}}) \quad \text{where } q \in A_i, \quad r \in A_j.$$

Thus

$$\mathbf{M}_{\text{GM}} := \{(i, j, O_{q\bar{r}}) \mid q \in A_i, \quad r \in A_j, \quad 1 \leq i, j \leq 3\}.$$

Proposition 64 (Conservative operator-basis lift). *Let*

$$\pi_{\text{QCD}}(i, j, O_{q\bar{r}}) = O_{q\bar{r}}.$$

Then

$$\pi_{\text{QCD}} \circ \Gamma_{\text{GM}} = \text{id}_{\mathbf{Q}_{\text{mes}}}.$$

Therefore the Grand Meson is a conservative lift of the QCD mesonic operator basis: it adds the three-axis pair label (i, j) while preserving the color-singlet QCD operator.

Remark 65 (Selector-relative completion property). Relative to a closed package selector $\hat{P}$, the object $\mathbf{GM}_{36}$ is used here as a selector-relative conservative channel-index lift of $\mathbf{Q}_{\text{mes}}$ satisfying the following conditions:

- (i) each $O_{q\bar{r}}$ has the unique generation-axis pair (i, j) determined by $q \in A_i$ and $r \in A_j$;
- (ii) forgetting the axis pair and package data recovers $O_{q\bar{r}}$;
- (iii) anti-face replacement and top-bearing ordinary readout produce the typed contradiction witnesses $\text{Contr}_{\bar{F}}$ and $\text{Contr}_{\text{top}}(-; \mathcal{R}_{\text{ch}})$ unless their residual channels close;
- (iv) no new color representation or color force is introduced;
- (v) the QCD quantum numbers, source Lagrangian, correlators, and CKM charged-current layer are preserved under forgetting.

This is a selector-relative labeling and closure property, not a category-theoretic universal property in an un-introduced category. If a later version defines such a category of completions, the required comparison map would have the form

$$\theta : \mathbf{GM}_{36} \longrightarrow \mathcal{X}$$

for a completion $\mathcal{X}$ equipped with its own recovery map $\pi_{\mathcal{X}} : \mathcal{X} \rightarrow \mathbf{Q}_{\text{mes}}$, with

$$\mathbf{GM}_{36}^{\text{ch,cl}} := \{(i, j, P_{q\bar{r}}) \in \mathbf{GM}_{36}^{\text{ch}} \mid P_{q\bar{r}} \in \text{Pol}_{q\bar{r}}^{\text{res}}\}$$

and a single dependent injection

$$\iota^{36} : \mathbf{GM}_{36}^{\text{ch,cl}} \rightarrow \mathbf{GM}_{36}, \quad \iota^{36}(i, j, P_{q\bar{r}}) := G^{(q\bar{r}, P)},$$

where all 36 slots are initialized by

$$G^{(q\bar{r}, P)}(a, \bar{b}) := \begin{cases} P, & (a, \bar{b}) = (q, \bar{r}), \\ \mathbf{0}_{a\bar{b}}[\text{buried}], & \text{otherwise.} \end{cases}$$

Choose a closed package selector

$$\widehat{P} : \mathbf{Q}_{\text{mes}} \rightarrow \text{Pol}^{\text{res}}, \quad \widehat{P}(O_{q\bar{r}}) \in \text{Pol}_{q\bar{r}}^{\text{res}}.$$

For top-bearing operators, $\widehat{P}$ selects only a weak-virtual closed package or a closed inactive residual package; it never selects an ordinary top-meson readout package. Then define

$$\widehat{\Gamma}_{\text{GM}}^{\widehat{P}}(O_{q\bar{r}}) := (i, j, \widehat{P}(O_{q\bar{r}})), \quad q \in A_i, \ r \in A_j.$$

Thus

$$\pi_{\mathcal{X}} \circ \theta \circ \iota^{36} \circ \widehat{\Gamma}_{\text{GM}}^{\widehat{P}} = \text{id}_{\mathbf{Q}_{\text{mes}}}.$$

Definition 66 (Grand Meson source Lagrangian). Write the quark-sector QCD Lagrangian as

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} G_{\mu\nu}^A G^{A\mu\nu} + \sum_{q \in F_6} \bar{q}(i\gamma^\mu D_\mu - m_q)q.$$

The Grand Meson source lift is

$$\mathcal{L}_{\text{GM}}[J] := \mathcal{L}_{\text{QCD}} + \sum_{i,j=1}^3 \sum_{\substack{q \in A_i \\ r \in A_j}} \sum_{\sigma} \left(J_{ij,\sigma}^{q\bar{r}} O_{q\bar{r}}^{\Gamma_\sigma} + \overline{J_{ij,\sigma}^{q\bar{r}}} (O_{q\bar{r}}^{\Gamma_\sigma})^\dagger \right).$$

Here $J_{ij,\sigma}^{q\bar{r}}$ is not a new color force. It is the closed Grand Meson source attached to the axis pair (i, j) and spin-parity sector σ . The Hermitian-conjugate term is included so that charged or otherwise non-Hermitian bilinears give a real Lagrangian density.

Proposition 67 (Lagrangian recovery). *Setting all Grand Meson sources to zero recovers QCD:*

$$\mathcal{L}_{\text{GM}}[0] = \mathcal{L}_{\text{QCD}}.$$

Forgetting the axis-pair labels sends the lifted source terms back to the ordinary mesonic source sector

$$J_\sigma^{q\bar{r}} := J_{ij,\sigma}^{q\bar{r}} \quad (q \in A_i, \ r \in A_j), \quad \sum_{q,r \in F_6} \sum_{\sigma} J_\sigma^{q\bar{r}} O_{q\bar{r}}^{\Gamma_\sigma}.$$

Thus the Grand Meson completes the mesonic Lagrangian source layer by adding axis-channel closure, not by altering the color-gauge core.

Definition 68 (Grand Meson quantum numbers). Recall the previously defined QCD quantum fingerprint $\nu_{\text{QCD}}(q, \bar{r}; \sigma)$. The Grand Meson lift is

$$\nu_{\text{GM}}(q, \bar{r}; \sigma) := (i, j, \nu_{\text{QCD}}(q, \bar{r}; \sigma)), \quad q \in A_i, \ r \in A_j.$$

Here σ may be J^{PC} for neutral channels; for charged or open-flavor channels it is the corresponding spin-parity sector without an unavailable charge-conjugation label.

Proposition 69 (Quantum-number recovery). *If $\pi_\nu(i, j, \nu) = \nu$, then*

$$\pi_\nu \circ \nu_{\text{GM}} = \nu_{\text{QCD}}.$$

Thus the Grand Meson completes mesonic quantum numbers by adding only the axis-pair quantum number (i, j) , while preserving charge, flavor, baryon number, color singletness, and the spin-parity sector.

Definition 70 (Grand Meson observables). For QCD mesonic operators $O_{q\bar{r}}^{\Gamma_\sigma}$, write the ordinary two-point observable as

$$C_{q\bar{r},s\bar{t}}^{\sigma\tau}(x-y) := \left\langle O_{q\bar{r}}^{\Gamma_\sigma}(x) (O_{s\bar{t}}^{\Gamma_\tau}(y))^\dagger \right\rangle.$$

The Grand Meson lift is the axis-closed observable

$$C_{ij,kl}^{\text{GM},\sigma\tau}(q\bar{r},s\bar{t};x-y) := (i,j,k,l, C_{q\bar{r},s\bar{t}}^{\sigma\tau}(x-y)),$$

where $q \in A_i$, $r \in A_j$, $s \in A_k$, $t \in A_l$.

Proposition 71 (Observable recovery). *Let $\pi_{\text{obs}}(i,j,k,l,C) = C$. Then*

$$\pi_{\text{obs}}(C_{ij,kl}^{\text{GM},\sigma\tau}(q\bar{r},s\bar{t};x-y)) = C_{q\bar{r},s\bar{t}}^{\sigma\tau}(x-y).$$

Thus masses, widths, and spectral poles read from ordinary QCD correlators are preserved, while the Grand Meson retains which axis-pair channel made the observable readable.

Definition 72 (Grand Meson conservation law). Let $\mathbf{a}_{ij}$ be the unit basis element of $\mathbb{Z}^{3 \times 3}$. For $q \in A_i$ and $r \in A_j$, define

$$K_{\text{QCD}}(q\bar{r}) := (Q_q - Q_r, e_q - e_r, \text{B}_{\text{ary}} = 0, \mathbf{1}_c)$$

and lift it to

$$K_{\text{GM}}(q\bar{r}) := (\mathbf{a}_{ij}, K_{\text{QCD}}(q\bar{r})).$$

Let $\mathcal{R}_{\text{axis}} \subseteq \mathbb{Z}^{3 \times 3}$ be the declared set of closed axis-residue records, and define

$$\text{AxisCl}(R) \iff R \in \mathcal{R}_{\text{axis}}.$$

Proposition 73 (Conservation-law recovery). *For an oriented mesonic process $\mathcal{P} = \sum_\alpha \eta_\alpha(q_\alpha \bar{r}_\alpha)$, ordinary QCD conservation is*

$$\sum_\alpha \eta_\alpha K_{\text{QCD}}(q_\alpha \bar{r}_\alpha) = 0.$$

The Grand Meson conservation condition is

$$R_{\text{axis}}(\mathcal{P}) := \sum_\alpha \eta_\alpha \mathbf{a}_{i_\alpha j_\alpha}, \quad \sum_\alpha \eta_\alpha K_{\text{GM}}(q_\alpha \bar{r}_\alpha) = (R_{\text{axis}}(\mathcal{P}), 0).$$

Thus QCD conservation is recovered by forgetting the axis residue R_{axis} . A closed Grand Meson process has $R_{\text{axis}} = 0$; otherwise the nonzero axis residue must satisfy

$$\text{AxisCl}(R_{\text{axis}}(\mathcal{P}))$$

rather than being ignored.

At this bookkeeping level, the mesonic flavor sector of QCD is represented compactly as the union of one-axis terms ($i = j$) and two-axis terms ($i \neq j$) among the three Grand Meson axes. The dynamics of color is not discarded; it is the singlet contraction already present in $O_{q\bar{r}}$. Channels touching t are retained as weak-virtual channels satisfying closure, not as ordinary top mesons.

11 Lorentz-Covariant Readout Condition

The Grand Meson must also avoid the classical aether failure. It is not a hidden medium with an absolute state of rest. It is a local six-face mediator with a 36-channel oriented body whose outputs must be readable without introducing an absolute frame. The nickname sometimes used for this condition is not a claim of Einstein's endorsement. It is only a shorthand for the Lorentz-covariant readout condition: no mesonic channel calculus may reintroduce an absolute hidden medium.

Axiom 74 (Lorentz-covariant readout: no absolute mesonic frame). No equation in the Grand Meson theory may depend on a privileged global rest frame of the mediator. Only resolved flavor channels, local channel closure, and observer-readable projections may be used. For a Lorentz transformation Λ , the flavor-axis labels (i, j) are internal labels and remain fixed, while spacetime arguments and tensor indices transform in the ordinary way:

$$(i, j, O_{q\bar{r}}^{\Gamma_\sigma}(x)) \mapsto (i, j, O_{q\bar{r}}^{\Gamma'_\sigma}(\Lambda x)).$$

Observable statements must therefore be Lorentz scalars or covariant tensor statements after forgetting the internal Grand Meson labels. The source layer must satisfy

$$\mathcal{L}_{\text{GM}}[J](x) \mapsto \mathcal{L}'_{\text{GM}}[J'](\Lambda x) = \mathcal{L}_{\text{GM}}[J](x),$$

and the lifted correlator must transform only through its spacetime arguments and tensor indices:

$$C_{ij,kl}^{\text{GM},\sigma\tau}(x-y) \mapsto C_{\text{GM}}'^{ij,kl,\sigma'\tau'}(\Lambda(x-y)).$$

Thus (i, j) are internal scalar labels, not aether-like spacetime degrees of freedom.

This axiom does not ban mediation. It bans absolute mediation. The Grand Meson is therefore not a return to the classical aether. It is a channel-resolved local mediator that must survive projection into closed observations.

12 Minimal Cover Relative to the Fixed Six-Flavor Input

Formally one may define an n -meson by

$$G_n := \{1, \dots, n\}, \quad F_{2n} := G_n \times H_2.$$

For a formal flavor set S , assume a typed pre-package family

$$\text{PrePol}_{a\bar{b}}^{S,\text{res}} \quad (a, b \in S),$$

and define the corresponding relative face by

$$\text{Face}_{a|S}^{S,\text{res}} := \prod_{b \in S} \text{PrePol}_{a\bar{b}}^{S,\text{res}} \times \prod_{c \in S} \text{PrePol}_{c\bar{a}}^{S,\text{res}}.$$

Then

$$\text{Mes}_n := \prod_{(i,\epsilon) \in F_{2n}} \text{Face}_{(i,\epsilon)|F_{2n}}^{F_{2n},\text{res}}.$$

The case $n = 1$ is the formal two-face analog of the General Meson; it becomes the General Meson after choosing the identification $F_2 \cong A_Y$. The case $n = 3$ becomes the Grand Meson after applying the observed-flavor identification $\Phi : F_6 \cong G_3 \times H_2$.

The reason to stop at $n = 3$ is not that higher n is impossible, and it is not a proof of why nature has exactly three generations. It is that the known six-flavor quark sector already factors as

$$F_6 \cong G_3 \times H_2.$$

Therefore $n = 3$ is the exact internal-generation cover of the present mesonic flavor sector:

$$\text{Mes}_3 \cong \prod_{(i,\epsilon) \in G_3 \times H_2} \text{Face}_{\Phi(i,\epsilon)|F_6}^{\text{res}}.$$

For $n > 3$, additional axes require additional readout structure. Such structure may be meaningful in a later theory, but it is not required for the present six-flavor sector. No order relation on readout power is used here; the claim is only minimality relative to the fixed six-flavor input.

Thus $n = 3$ is the minimal complete cover for the present Grand Meson input: complete because it covers the known three-generation quark sector, minimal because no fourth axis is introduced without a corresponding standard-model flavor input. This is not a theorem explaining why nature has three generations. It is the closure statement for the currently used six-flavor quark input; if the input sector changes, the update procedure must be rerun rather than silently extended.

13 Closed Transition

Grand Meson robustness is not the absence of change. It is the closure of every changed face or body channel. The body version was defined in the Grand Fuse section. The face version is as follows. Let a proposed face transition symbol be written

$$u : g \rightarrow g'.$$

It carries source and target skeletons

$$s(u) = g, \quad t(u) = g'.$$

Define its changed-face support by

$$\Delta_{\text{face}}(u) := \{q \in F_6 \mid s(u)(q) \neq t(u)(q)\}.$$

Definition 75 (Grand Meson transition space). The face transition space is

$$\text{Mor}_{\text{GM}}^{\text{face}} := \{u : g \rightarrow g' \mid \Delta_{\text{face}}(u) \subseteq \text{RCl}_{\text{face}}(u)\}.$$

The contradiction predicate for a proposed transition symbol is

$$\text{Contr}_{\text{tr}}(u) := \Delta_{\text{face}}(u) \setminus \text{RCl}_{\text{face}}(u) \neq \emptyset.$$

Proposition 76 (Transition block). *If $\text{Contr}_{\text{tr}}(u)$, then*

$$u \notin \text{Mor}_{\text{GM}}^{\text{face}}.$$

Consequently a channel-boundary crossing without closure does not name an operation inside the Grand Meson theory; it is a contradiction witness outside the transition space.

Proof. By definition, $u \in \text{Mor}_{\text{GM}}^{\text{face}}$ requires $\Delta_{\text{face}}(u) \subseteq \text{RCl}_{\text{face}}(u)$. The predicate $\text{Contr}_{\text{tr}}(u)$ states that $\Delta_{\text{face}}(u) \setminus \text{RCl}_{\text{face}}(u)$ is nonempty. These conditions are inconsistent, so the proposed symbol cannot inhabit $\text{Mor}_{\text{GM}}^{\text{face}}$. $\square$

Definition 77 (Physical well-formedness). A physically well-formed Grand Meson transition must satisfy

$\text{Cons}_{\text{QCD}}(u) \iff$ the QCD conservation vector of the transition carried by u is zero,

$\text{Lag}_{\text{GM}}(u) \iff$ the transition carried by u has a declared Grand Meson source-layer term,
and

$\text{Obs}_{\text{GM}}(u) \iff$ the transition carried by u has a declared observable readout or correlator.

Then

$$\text{PhysWF}(u) := (u \in \text{Mor}_{\text{GM}}^{\text{face}}) \wedge \text{Cons}_{\text{QCD}}(u) \wedge \text{Lag}_{\text{GM}}(u) \wedge \text{Obs}_{\text{GM}}(u).$$

Here Cons_{QCD} means that the conservation condition closes, Lag_{GM} means that the transition has a defined Lagrangian-source layer, and Obs_{GM} means that an observable readout exists.

Equivalently,

$$\text{non-transition} = \text{unclosed discontinuity}.$$

The discontinuity itself is not the contradiction. Without channel closure, the corresponding transition symbol has no place in the typed Grand Meson transition spaces.

14 Conclusion

The Grand Meson is not proposed as an omnipotent meson. It is proposed as a closed $n = 3$ channel-completion calculus for mesonic sectors:

$$\text{Grand} = \text{Grand}_{\text{comp}} = \text{Im}(\text{skel} : \text{Body}_{36}^{\text{cl}} \rightarrow \text{Grand}^{\text{raw}}), \quad \text{GM}_{36} = \text{Body}_{36}^{\text{cl}}.$$

It conservatively recovers the General Meson by a surjective projection, replaces missing channels with resolved flavor channels, avoids an absolute aether frame, and treats robustness as channel closure rather than immobility. Its backward compatibility is therefore responsible: extra axes may be fused into the older face only with residual closure, while unclosed fission reduces to a typed Grand Meson contradiction and has an empty transition class.

The path is therefore narrow. Begin with the $u \leftrightarrow d$ eye that already works. Open $c \leftrightarrow s$ and $t \leftrightarrow b$ without breaking it. Stop at $n = 3$ until a future theory can make higher axes readable without turning them into a burden of infinite explanation.

Non-Technical Appendix: Observer-Side Fission

This note is not used in any theorem above. It records a philosophical reading of the downgrade discipline. Let a concept or entity X be represented by a closed Grand Meson state g_X . A thought experiment may introduce a proposed fission symbol $d : g_X \Downarrow m_X$ and later reclose the discussion by fusion. This does not damage the technical calculus when the whole argument remains inside the typed transition spaces. If, however, the proposed fission leaves a contradiction witness

$$\text{Contr}_{\text{dow}}(d),$$

then the symbol has already left the Grand Meson transition space. As an observer-side convention, a later observer class $\mathcal{O}$ may record a closed reading and a fission-intent reading,

$$r_{\text{closed}}, r_{\text{intent}} \in \text{Read}_{\mathcal{O}}(X), \quad r_{\text{closed}} \neq r_{\text{intent}}.$$

This is an interpretive appendix, not a consequence of Lorentz covariance or of the Lorentz-covariant readout condition.

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